Highlights

Calibrated Decisions at Scale: Converting Police Crash Narratives into Probabilistic Crash Variables with a System One Model (Jev)

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- A typed decision model codes crash narratives at scale with a measured cost model
- Its probabilities rank well but overstate prevalence until recalibrated on labels
- A two-decimal output grid puts a floor on calibration error for rare variables
- The human-review budget must be defined over flagged records rather than all records
- Adding calibrated narrative variables changes which factors a safety diagnosis ranks

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Calibrated Decisions at Scale: Converting Police Crash Narratives into Probabilistic Crash Variables with a System One Model (Jev)

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ABSTRACT

Crash datasets that carry an investigator narrative hold information the coded fields omit. Coding those narratives at scale has been blocked by three obstacles. Frontier large language models are costly at that scale, their generated text cannot be verified, and no rule says how much output a human must check. This paper formulates narrative coding as gated, typed decisions answered by Jev, a System One model that returns probabilities over analyst-defined options and generates no text. A screen covered 499,500 Texas narratives and 195,857 were coded with a 27-question schema. Cost is governed by schema size rather than narrative length. The probabilities are audited against coded fields and against 2,416 blinded human judgments drawn under a stated sampling design. Two frontier large language models are benchmarked on the same records. Against human labels the typed model attains an F_1 of 0.908. One frontier model gains 0.059 and the other is indistinguishable from it. Calibration varies by model rather than by paradigm, so each model must be audited. Recalibration on the same labels reduces calibration error by a factor of 3.3. Agreement with coded fields understates fidelity to the narrative by a median of 0.26 in kappa. A resolution-floor bound covers any model that reports probabilities on a discrete grid. A review budget over flagged records gives the records a human must read per variable and per year. Adding the calibrated variables to the coded fields raises the injury and fatal crashes attributed to nine factors by 10,747 per year.

1. Introduction

The investigator narrative is the richest field in a crash database and the one that safety analysis uses least. The Texas Crash Records Information System (CRIS) holds 5,109,746 crash records for 2017 to 2025, of which 5,018,080 carry a narrative longer than 40 characters, which is about 557,564 new narratives every year. The coded fields of a crash report are fixed by the design of the form, so they record what the form asks and nothing else. The narrative is where the officer writes that the driver had a seizure, fell asleep, hydroplaned, was pregnant, was chased by another motorist, or lost a wheel. Several of these events have no coded field at all, and others have a field that is filled inconsistently. Text mining of narratives has already recovered hydroplaning crashes that no field records (Das et al., 2020), work-zone crashes that the coded fields misclassified (Sayed et al., 2021), agricultural crashes hidden in general categories (Kim et al., 2021), and speeding designations that the coded flag missed (Fitzpatrick et al., 2017). Each of those studies treated the narrative as a record of circumstances that the coded fields do not carry, which is the premise of the present work. What a narrative supplies is the officer's written account, so the quantity this paper measures is what the narrative states rather than what occurred, and every claim below is framed that way.

Three obstacles have kept narrative coding out of routine agency practice despite this evidence, and the first of them is cost. Running a frontier language model over a state's full narrative corpus is expensive enough that it is not done, so studies code a few thousand records and extrapolate. The largest recent benchmark of frontier models against official crash coding used some four thousand matched Arkansas fatal crashes (Bharati et al., 2026), and the Kentucky secondary-crash audit worked on a single variable in some sixteen thousand manually reviewed narratives (Zhang and Chen, 2026). A method that cannot be afforded at population scale cannot replace or supplement coded fields at population scale, whatever its accuracy on a sample.

The second obstacle is the verification of generated output. A generative model asked to return labels or a JSON object can invent option values, omit fields, and answer a question that was not asked, so every output needs a parser and a validity check. When such a model is asked how confident it is, the stated number is a piece of generated text

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rather than a measured probability. Whether that number is calibrated depends on how it is elicited (Kadavath et al., 2022; Tian et al., 2023), and verbalized confidence tends toward over-confidence (Xiong et al., 2023). A coding pipeline whose confidence values cannot be trusted forces an agency either to check everything or to check nothing.

The third obstacle, and the one this paper is built around, is that no published study tells a safety office how much human checking a narrative-derived variable needs. Agreement with coded fields is the usual evaluation, and it answers a different question. A classifier that is right most of the time does not tell an analyst which records to open. A probability that is right as often as it claims to be does, because a target precision can then be converted into a share of flagged records that a person must read. That conversion is only valid if the probabilities are calibrated, and calibration is a property that has to be measured against an external reference rather than assumed from a vendor's description.

The situation changed with the arrival of typed decision models. In September 2026 TypeSafe released Jev, which its developer describes as a System One model (TypeSafe, 2026a,c). Such a model takes a state object and a set of typed questions and returns, for each question, an answer from a fixed option set together with a probability, in one pass and with no generated text. The developer states that the training objective optimizes probabilities against outcomes and prices the model at $0.042 per million input tokens with no charge for output (TypeSafe, 2026b). If those probabilities are calibrated, the review budget of a narrative-derived variable becomes a computable quantity and the cost of computing it over a whole state corpus is measured in hundreds of dollars. Whether they are calibrated is an empirical question on which the developer has published no metrics, and it is the question this paper answers independently.

No published work combines the four elements that such an answer requires. None uses a non-generative typed-decision model for crash-narrative coding. None audits the calibration of narrative-derived probabilities variable by variable against both a large administrative reference and a human reference with known sampling weights. None converts calibration into a human-review budget defined over the records an agency would actually examine. And none does these things at the scale of a state corpus with a cost model measured rather than projected. The nearest studies evaluate generative models on attributes the database already codes and report agreement (Bharati et al., 2026; Zhang and Chen, 2026), which is a sound design for what it measures but leaves the calibration and the review budget untouched.

Those gaps define the four research questions that this paper answers. The first asks whether a typed decision model can code narratives at the scale of a state corpus with a schema written by a safety analyst, and at what cost, so that large-scale coding can replace designs that code a sample and extrapolate. The second asks whether the probabilities such a model returns are calibrated, measured first against the coded fields and then against human judgment of the same narratives, and whether the answer holds for frontier generative models run on the same records. The third asks how much human review a narrative-derived variable needs once its calibration is known, expressed per variable in records an agency must read each year. The fourth asks what changes in a safety diagnosis when calibrated narrative variables are added to the coded contributing-factor fields, which is the question that decides whether any of the rest is worth an agency's attention. The paper answers the first affirmatively and reports the executed sizes rather than a projection. For the second it finds that the typed model and one frontier arm are not calibrated as delivered, while the other frontier arm is. It answers the third with a budget defined over flagged records and certified out of fold, and the fourth by measuring the injury and fatal crashes that the coded fields miss and the narratives supply.

This paper makes five connected contributions to the coding of crash narratives. It first formulates narrative coding as a set of gated, typed decisions, in which presence questions open detail questions and named entities are selected from a closed option set rather than generated. The effect of each design rule is measured instead of asserted, including a paired before-and-after test of an exclusion rule on a fixed sample. The paper then runs that decomposition at scale, screening 499,500 Texas narratives with a 10-question schema and coding 195,857 of them with the full 27-question schema, at a measured cost of $0.154 per thousand narratives. Applied to the whole corpus of 5,018,080 narratives at that rate the projected cost is $774, and the corpus itself was not coded. The fitted cost model shows that spend is governed by the size of the question schema rather than by the length of the text, which reverses the premise behind two-stage screening designs.

The third contribution is an independent calibration and selective-prediction audit of every variable, against coded CRIS fields as a large agreement reference and against a cell-stratified human reference set with calibration weights. That second reference measures fidelity to the narrative rather than the facts of the crash, because a coder is asked what the narrative says and not what probably happened. The audit finds that the model returns probabilities on a discrete two-decimal grid whose floor exceeds the base rate of several target variables, which places a floor on the attainable calibration error that no improvement in ranking can remove. The fourth contribution is a reformulation

of the review budget. Accuracy-based selective risk carries little information when the positive class is rare, because abstaining on nothing already satisfies conventional risk targets. The budget is therefore defined over flagged records at a target precision, certified on narratives that did not select the threshold, and reported per variable per year. The fifth contribution joins the audit to a transportation outcome. It measures how the injury and fatal crashes attributed to each contributing factor change when the calibrated narrative variables are added to the coded fields, and the rate at which coders confirm narrative-only flags decides which of those changes can be believed. Two frontier generative models are run on the same human-labeled records as an accuracy comparison, and the schema, the runner, the metric code and every aggregated output behind a figure or table are released.

The rest of the paper proceeds as follows. Section 2 places the work in the literatures on narrative coding, calibration and selective prediction, and typed decision models. Section 3 describes the corpus, the schema, the reference sets and the evaluation, and states the quantities the analysis computes as numbered equations. Section 4 reports the population run, the calibration audit, the review budgets and the comparison with keyword rules and frontier models. Section 5 compares the findings with the closest published studies, and Section 6 summarizes the contributions, states the boundaries of the study and sets out the work that follows from them.

2. Background

2.1. Crash narratives and narrative coding in safety research

Work on crash narratives has moved from keyword rules, through topic models and fine-tuned encoders, to zero-shot prompting of large language models, and the coded field has served as the reference at every stage. Rule-based and feature-based methods identified work-zone crashes that the coded fields misclassified (Sayed et al., 2021), agricultural crashes that a curated keyword list could recover from general categories (Kim et al., 2021), and contributory factors beyond the single code a form permits (Zou et al., 2024). They also identified hydroplaning involvement, which no CRIS field records and which interpretable models could associate with vehicle characteristics only after the narrative supplied the label (Das et al., 2020). Topic modeling exposed latent themes across a decade of fatal crash narratives (Kwayu et al., 2021), and fine-tuned encoders extended the approach to pedal misapplication (Bareiss et al., 2021) and to severity classification on Texas narratives (Oliaee et al., 2023). The same progression occurred in adjacent domains, with construction-accident narratives classified by support vector machines and other classical learners (Goh and Ubeynarayana, 2017) and railway-accident narratives by deep networks (Heidarysafa et al., 2018). Two properties recur across this literature and carry into the present design. The narrative is reliable enough to serve as the standard against which a coded field is judged, as in the speeding-designation review of Fitzpatrick et al. (2017), and it exposes under-reporting in the coded fields themselves, which Arteaga and Park (2025) used a language-model framework to quantify.

Two recent studies bracket the current state of practice with language models and together motivate this paper. Bharati et al. (2026) evaluated six frontier language models against official Arkansas fatal-crash coding on some four thousand matched crashes over six attributes the database already codes. They found that a keyword-rule baseline performed comparably to the best model and that differences across attributes exceeded differences across models, from which they recommended attribute-specific evaluation and human review. This paper takes that recommendation as its starting point, evaluates every variable separately, and quantifies the human review the recommendation calls for. Zhang and Chen (2026) audited a single variable, secondary-crash involvement, in Kentucky narratives and showed that a fine-tuned transformer beat prompted language models at a fraction of their cost, which sets the cost bar a new method has to clear. Earlier comparative work on ChatGPT, BARD and GPT-4 was qualitative and small (Mumtarin et al., 2023), and more recent frameworks generate explanations and causal chains rather than audited probabilities (Zhen and Yang, 2025; Su et al., 2026). Fine-tuning a language model to infer hazardous driver actions from narratives improved on inconsistent manual coding and produced probabilistic outputs, but it required labeled training data and reported no calibration audit (Chen et al., 2026). Instruction-tuned models have been prompted for vulnerability indicators in police text without a confidence audit (Relins et al., 2025). A recent systematic review of natural language processing applied to crash reports covers severity, duration and causation and reports no calibrated, selective coding (Bazdar et al., 2026). Outside transportation, language models have been evaluated as text annotators (Gilardi et al., 2023; Ding et al., 2023; Ziems et al., 2024) and for structured extraction from scientific prose (Dagdelen et al., 2024), and those studies report agreement with human labels rather than calibration. The clinical-text literature, which is several years ahead on validating model-extracted variables, reaches a consistent verdict that carries over here. Agreement with expert adjudication is high on a binary progression question assessed from radiology reports (Kristjánsson et al.,

2026). It is also high on the presence and grade of adverse events extracted from oncology notes but degrades on their finer attributes (Guillot et al., 2026), and it degrades on the finer grades of postoperative complications classified from discharge letters (Warmer et al., 2026). That literature also supplies three cautions that shaped the schema design. Extraction performance depends heavily on how the prompt is constructed (Wu et al., 2026), prompt architecture induces order and label biases that survive an instruction that the options were shuffled (Brucks and Toubia, 2025), and the largest model is not always the right one, since fine-tuned encoders beat decoder-only models on overdose-death classification (Funnell et al., 2026). A typed interface with a fixed option set narrows the surface those two cautions apply to, because the model selects from an option set the analyst fixes and returns no free text, so no answer can be invalid and none has to be parsed. Whether the option order or the criteria wording still moves the returned probabilities is a separate question, and this study does not test it, so the claim made here is about output validity rather than about the absence of order or wording effects.

The injury-surveillance literature on auto-coding narratives is the same problem in a neighboring field and is more mature on two points that this paper needs. It confronted the division of labor between machine and human early, with Bayesian classifiers coding narratives into cause groups (Lehto et al., 2009) and human-machine ensembles routing the narratives coded with low confidence to manual review (Marucci-Wellman et al., 2017). The same group developed semi-automated coding of short injury narratives from large administrative databases (Marucci-Wellman et al., 2015), and Naive Bayes coding was applied to workers' compensation claims and to near-miss narratives from the fire service (Bertke et al., 2012; Taylor et al., 2014). It also confronted rare categories explicitly, asking whether more training data or filtering rescues them (Nanda et al., 2018) and comparing auto-coding methods on the same causation task (Bertke et al., 2016), with factorization models as one answer (Chen et al., 2015). Early work showed that computerized coding of narrative text improves surveillance when a human remains in the loop (Wellman et al., 2004), and a systematic review surveys the machine-learning approaches to that surveillance text (Vallmuur, 2015). What that line did not do is measure whether the confidence that governs the routing is calibrated, which is what distinguishes a review threshold from a review budget.

The coded fields that narrative studies use as their reference carry documented error, and the direction of that error matters for the audit reported here. The validity of police-reported crash data has been examined since Shinar et al. (1983), and Austin (1995) showed that mistakes in the locational variables of accident records can be identified by comparing the report with accurate highway data from other sources. Under-reporting of crashes and injuries to police is large and uneven across countries, severities and road users (Elvik and Mysen, 1999; Alsop and Langley, 2001; Amoros et al., 2006). Linked police and hospital records show that the police file misses injured people it should contain (Watson et al., 2015; Janstrup et al., 2016), and police-reported injury codes are an inadequate proxy for the injury severity that in-depth investigation records (Farmer, 2003). Under-reporting also biases the parameters of severity models (Yamamoto et al., 2008; Ye and Lord, 2011), and a review of crash data quality for road-safety research lists incompleteness of contributing-factor fields among its open problems (Imprialou and Quddus, 2019). The common pattern across these studies is omission, because a coded field fails to record a factor far more often than it records one that did not occur, and that is the one-sided error structure that Section 3.4 formalizes. De-identification is the last concern the narrative literatures share. Automated removal of identifiers from free text has a long record in clinical narratives (Uzuner et al., 2007; Neamatullah et al., 2008; Meystre et al., 2010; Stubbs et al., 2015), and Ma et al. (2026) address it for crash narratives with a local agentic workflow that reaches high recall. This paper applies a simpler regular-expression pass and a model-based screen, and reports residual-identifier rates rather than claiming removal.

2.2. Calibration, selective prediction, and label-efficient evaluation

A probabilistic classifier is calibrated when, among the cases it assigns probability p, a fraction p are positive, and the tools for assessing that property come from forecast verification. The Brier score (BRIER, 1950) is a strictly proper scoring rule (Gneiting and Raftery, 2007), and its decomposition into reliability, resolution and uncertainty (Murphy, 1973) separates calibration from discrimination. Subjective probability forecasts of weather were the first to be assessed this way (Murphy and Winkler, 1977), and DeGroot and Fienberg (1983) formalized the comparison of forecasters by calibration and refinement, which is why the forecasting literature frames the target as maximizing sharpness subject to calibration (Gneiting et al., 2007). Reliability diagrams make calibration visible, and their binning has been put on a firmer footing by consistency bands (Bröcker and Smith, 2007), by the decomposition of proper scores (Bröcker, 2009), and by binning-free estimators that make the diagram reproducible (Dimitriadis et al., 2021; Gneiting and Resin, 2023; Dimitriadis et al., 2024). In machine learning the same quantity is summarized by expected calibration

error (Pakdaman Naeini et al., 2015), which is sensitive to its binning (Nixon et al., 2019), whose estimation properties were examined by Vaicenavicius et al. (2019), and whose verified estimation requires discretized outputs (Kumar et al., 2019). Modern neural classifiers are usually poorly calibrated (Guo et al., 2017), which is why calibration has to be measured rather than assumed. Post-hoc recalibration by logistic scaling (Platt, 2000), isotonic regression (Zadrozny and Elkan, 2002), and their comparison across learners (Niculescu-Mizil and Caruana, 2005) remain the standard remedies, extended by beta calibration (Kull et al., 2017) and Dirichlet calibration (Kull et al., 2019).

The clinical prediction-model literature adds the operational reading that this paper adopts. Calibration is assessed by a logistic regression of the outcome on the logit of the prediction, whose slope and intercept were introduced by Cox (1958), and by the test statistic of Spiegelhalter (1986). Van Calster et al. (2016) defined a hierarchy of calibration from the mean level through the slope to moderate and strong calibration, and Van Calster et al. (2019) argue that poorly calibrated algorithms are misleading and potentially harmful for decision-making. The quantity that matters for a decision is the net benefit at the threshold actually used (Vickers and Elkin, 2006; Steyerberg et al., 2010), and the textbook of Steyerberg (2009) treats recalibration as a routine step in model updating. This paper carries that view to a review budget, where the decision threshold is a target precision and the benefit is the share of flagged records that need not be read.

Selective classification supplies the decision-theoretic frame for abstention. The optimum error-reject trade-off was characterized by Chow (1970), and the modern formulation traces selective risk against coverage as the abstention threshold moves (El-Yaniv and Wiener, 2010; Geifman and Yaniv, 2017). Theoretical guarantees exist for pointwise-competitive selection (Wiener and El-Yaniv, 2015), hinge-loss and support-vector formulations of the reject option (Bartlett and Wegkamp, 2008; Wegkamp and Yuan, 2011), and a general learning theory of rejection (Cortes et al., 2016). Deep networks have been given an integrated reject option (Geifman and El-Yaniv, 2019), with the maximum softmax probability as the baseline confidence score (Hendrycks and Gimpel, 2016). That literature was developed for balanced problems, and its risk targets assume them. Narrative-derived safety variables are rare, and the precision-recall literature has long observed that error rates dominated by the negative class conceal what happens on the positive class (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015). Section 3.4 shows that the standard formulation becomes vacuous in that regime and defines the budget over flagged records instead.

Label-efficient evaluation is the last ingredient of the design. The Horvitz–Thompson estimator (Horvitz and Thompson, 1952) makes population estimates from a sample with unequal inclusion probabilities, with the design-based variance theory of survey sampling behind it (Kish, 1965; Särndal et al., 1992) and the bootstrap as a general interval method (Efron and Tibshirani, 1993). Active evaluation applies the same idea to classifier assessment, choosing which items to label so that a performance estimate is precise at a fixed labeling cost. It has been developed for F-measures (Sawade et al., 2010), for classifiers at web scale (Bennett and Carvalho, 2010; Katariya et al., 2012), for entity resolution (Marchant and Rubinstein, 2017), for rare categories (Poms et al., 2021), and for model evaluation in general (Kossen et al., 2021). The reference set of this paper follows that design, stratifying on the model's own probability so that the middle of the range, where calibration is informative and cases are scarce, is estimable at all. Agreement between coders is measured with Cohen's kappa (Cohen, 1960) on the scale of Landis and Koch (1977), with the cautions of McHugh (2012), and exact binomial intervals (CLOPPER and PEARSON, 1934) and the paired test of McNemar (1947) complete the toolkit.

2.3. Decision models without text generation and the calibration of language-model confidence

The model audited here belongs to a class its developer calls System One models (TypeSafe, 2026c). Such a model takes an unstructured state and a set of typed questions and returns, for each question, an answer from the option set the analyst defined together with a probability, in a single parallel pass and with no generated text. The developer describes the training objective as reinforcement learning for calibrated decisions and states that the probabilities are optimized against outcomes, that the model never makes a type error, and that higher confidence means higher accuracy (TypeSafe, 2026a). Three question types exist, a binary presence question, a choice among named options, and an ordered score, and the model is priced per input token with output free (TypeSafe, 2026b). These claims are quoted here rather than adopted. The documentation states no numeric precision, granularity or bounds for the returned probabilities, and the developer has published no calibration metrics, so the output grid reported in Section 4.1 and the calibration reported in Section 4.2 are measurements of this paper rather than documented properties.

The contrast with generative models is in how a probability is obtained, and the distinction has three levels that are easily conflated. A generative model does produce token likelihoods, which are well defined and describe the words it selected, but a label's probability is not one of them and the likelihood of the token "yes" is not the model's

belief that the answer is yes. A second level is self-evaluation, in which the model is asked whether its own answer is correct. A third is verbalized confidence, in which the model is asked to state a number. The second and third have been studied intensively, and the third is what the frontier arms of this paper supply. Pre-trained transformers were found to be reasonably calibrated in domain out of the box (Desai and Durrett, 2020), and generative language models answering factual questions were found to be poorly calibrated (Jiang et al., 2021). Large models can predict whether they know an answer when asked directly, but the property degrades with fine-tuning (Kadavath et al., 2022), and models can be trained to state uncertainty in words (Lin et al., 2022). Conversational agents were over-confident until linguistically calibrated (Mielke et al., 2022), and eliciting usable confidence from a model tuned with human feedback requires deliberate strategies (Tian et al., 2023). An empirical comparison of elicitation methods found that verbalized confidence tends to be over-confident (Xiong et al., 2023). Semantic entropy over sampled answers has been proposed as an alternative (Kuhn et al., 2023), and a survey of the field catalogs the confidence-estimation and calibration methods proposed to date together with their open challenges (Geng et al., 2024). Zhou et al. (2024) report that larger and more instructable models become less reliable in the sense that they answer more and are wrong more often on hard items. A typed interface that returns a probability directly sidesteps elicitation, which is exactly why its calibration has to be audited rather than inferred.

Structured-output and cascade designs are the engineering context. Constrained decoding guarantees that a generated string matches a grammar (Willard and Louf, 2023), which removes type errors but says nothing about confidence, and cascades route inputs to a cheaper model first and escalate on low confidence (Chen et al., 2023b), which presupposes that the confidence is meaningful. Selective prediction for language models follows the same logic, abstaining on domain shift (Kamath et al., 2020), across settings (Varshney et al., 2022), and after self-evaluation (Chen et al., 2023a). The vendor's own examples of confidence-gated fallback use small demonstrations without an external reference (TypeSafe, 2026c).

Taken together, the three literatures leave a specific gap. Narrative coding has reached population scale with rules and sample scale with generative models, and it validates against coded fields whose error is one-sided. Calibration and selective prediction supply the estimators and the decision frame, and both were developed for continuous outputs and for classes that are not extremely rare. Neither assumption is a flaw in that literature, and neither holds here, so each has to be replaced with something specific to this setting. A discrete output grid admits an exact calibration error and imposes a bound on how small that error can be, which is a property of the model's interface rather than of its skill. A rare positive class makes an error-rate target uninformative and calls instead for a measure of the workload a precision requirement implies. Typed decision models return probabilities cheaply and without elicitation, but no independent measurement of those probabilities exists on any safety corpus. This paper fills that gap on four fronts. It audits a typed decision model against both an administrative and a human reference, and it derives the grid bound and applies the exact estimator that bound permits. It defines the review budget over the records an agency would read, and it measures what the resulting variables change in a safety diagnosis.

3. Data and methods

Figure 1 gives the shape of the pipeline before the parts are defined. A crash narrative enters as program state, a typed schema of 16 presence, 8 categorical and 3 ordinal questions is put to the model in one call, and the model, which is fixed and never trained here, returns a probability per question. Those probabilities form a table in which every cell carries its own uncertainty, and the only fitted object in the whole pipeline is the one-dimensional recalibration map that turns the returned probabilities into calibrated ones. Two references, drawn below the pipeline, decide what a probability means. The coded CRIS fields are compared against, at the scale of the random stratum, but never fitted on, and the human reference set supplies the labels on which the recalibration map is fitted and on which accuracy is measured. The right-hand block of the figure is the deliverable of the audit, the share of flagged records a person must open to reach a target precision, which the results report as 0.8% at the 90% target. Table 1 collects the notation used in this section, and Table D.1 in Appendix D maps every numbered equation to the script and result file that computes it. The section first describes the corpus and its de-identification, then the schema and the two-stage coding design with its cost model, then the three reference sets against which the model is scored, and finally the evaluation quantities.

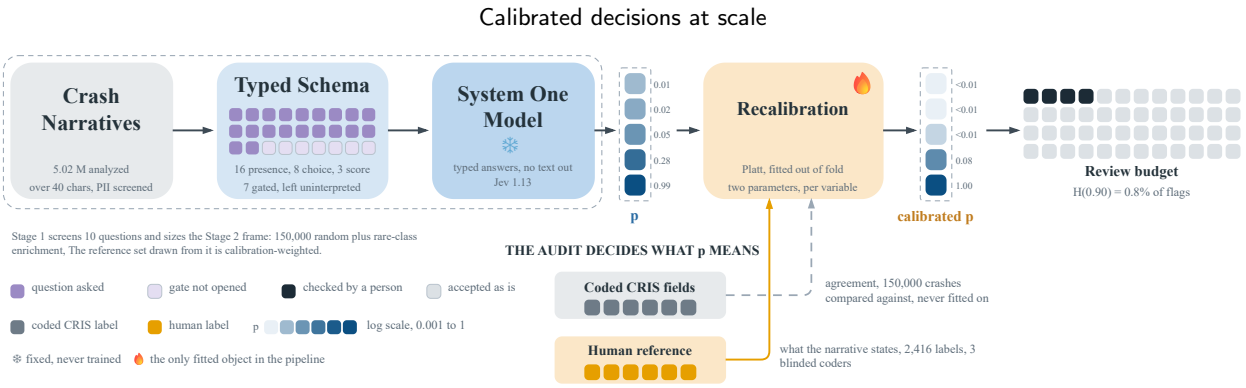


Figure 1: The framework. The pipeline runs from left to right, starting with the corpus of 5,018,080 screened narratives and the typed schema, a grid of questions in which filled squares are asked of every narrative and pale squares wait on a presence question. The snowflake marks the model as fixed and never trained, the flame marks the recalibration map as the only fitted object, and the grid at the right is the review budget, its dark squares the records a person must check. The coded CRIS fields (gray) join by a dashed line because they are only compared against; the human reference set (orange) joins by a solid line because it supplies the labels the map is fitted on.

Table 1
Notation. Inclusion probabilities carry one index and answer distributions carry two, and the target precision carries a star, so the three uses of π do not collide.

Symbol	Meaning
n_i, $i = 1, \dots, N$	narrative i of the analyzed corpus of size N
$\mathcal{V} = \mathcal{B} \cup \mathcal{C} \cup \mathcal{S}$	variables, partitioned into presence, categorical and ordinal sets
O_v, $o_v^{\varnothing}$	option set of a categorical variable and its no-match option
$p_{iv} \in [0, 1]$	returned probability that presence variable v holds in narrative i
$\pi_{iv} \in \Delta(O_v)$, $\hat{o}_{iv}$	returned distribution over options and its mode
$c_{iv} = \phi(\pi_{iv})$	returned confidence attached to a categorical or ordinal answer
$g(v)$, τ_g	gate of a dependent variable and the gate threshold
λ_v	leakage rate of a gated variable
$y_{iv} \in \{0, 1\}$, y'_{iv}	true label and coded-field label
$\hat{y}_{iv} = \mathbb{1}[p_{iv} > \tau]$	decision at threshold τ
θ	abstention threshold on the confidence score, distinct from τ
w_i	calibration weight of a pair in the human reference sample
a_{iv}	probability that variable v was assigned to narrative i for labeling
ρ	base rate (prevalence) of a variable
G, δ, w_0	attainable output grid, its smallest positive value, and the mass on zero
κ_{iv}	confidence score used for abstention
$C(\cdot)$, $R(\cdot)$	coverage and selective risk
π^*, $H(\pi^*)$	target precision and the review budget at that precision
k, ℓ_i	questions per call and narrative length in characters

3.1. Corpus and de-identification

The corpus is the crash-level narrative file of the Texas Crash Records Information System for the years 2017 to 2025, held under the authors' data agreement with the Texas Department of Transportation. It is joined to the crash-unit-person extract of the same system by crash identifier. The narrative file holds 5,109,746 crash records, and after removing records whose narrative is empty or has 40 characters or fewer, 5,018,080 narratives remain and form the analyzed corpus. The median narrative has 378 characters, the ninety-fifth percentile has 935, and the ninety-ninth has 1427, so a typical narrative contributes about 89 input tokens to a model call. Narratives are written either in mixed case or in capitals, in a ratio that is stable across years and serves as a proxy for reporting agency (Figure 2c). Narrative availability changed sharply over the period, because the share of crash records with an empty narrative falls from 8.24%

in 2017 to 0.00% in 2025 (Figure 2a). Any analysis that pools years without conditioning on narrative availability underrepresents the early period, and prevalence trends estimated from narratives alone are confounded with this reporting change. Every prevalence reported here is therefore stated on the analyzed corpus with its denominator explicit.

Names and telephone numbers were replaced with placeholders by the data owner before the file was received, and the identifiers that survive that pass were measured over all 5,018,080 narratives with the regular expressions released with the code. Date strings occur in 3.13% of narratives, long digit runs in 1.93%, case or report numbers in 0.77%, date-of-birth tokens in 0.29%, and driver-license or identification numbers in 0.20%. Titled personal names, such as an officer's rank followed by a surname, match in 4.65% of narratives, though most of these name the reporting officer rather than an involved party. A second-pass regular-expression redaction replaces each of these patterns with a placeholder in any copy of a narrative used for display. A model-based residual-identifier screen, which asks a single presence question about any unreplaced name, date of birth, license number, address or telephone number, is applied before any narrative is shown to a person outside the research team.

The sequence in which those three protections were applied differs by recipient, and it is stated here exactly rather than summarized. The data owner's redaction of names and telephone numbers applies to every narrative and preceded everything else. The typed-model runs, both the first-stage screen of 499,500 narratives and the full coding of 195,857, sent that owner-redacted text and nothing further, so the second-pass regular-expression redaction and the model-based screen were not applied before them. Those two further protections were applied to the 640 candidates of the reference frame, and only to them, before any narrative reached a human coder or either frontier vendor. 137 candidates were withheld on the screen's account and replaced. The consequence is that the human coders and the two frontier vendors received twice-redacted and screened text, while the typed-model service received the owner-redacted text at the rates of residual identifiers reported above. That asymmetry follows from the order in which the work was done rather than from a judgment that one recipient needs less protection, and an agency repeating this work should apply the screen before the first model call. No narrative is reproduced in this paper, and every example of narrative text is synthetic.

3.2. Schema design and two-stage coding

Narrative coding is formulated as a set of typed decisions over the narratives n_i and the variable set $\mathcal{V}$, which is partitioned into presence variables $\mathcal{B}$, categorical variables $\mathcal{C}$ with an explicit no-match option $o_v^{\varnothing} \in O_v$, and ordinal variables $\mathcal{S}$ with levels $0, \dots, L_v$. For a presence variable the model returns a probability $p_{iv} \in [0, 1]$ that the variable holds. For a categorical variable it returns a distribution $\pi_{iv} \in \Delta(O_v)$ with mode $\hat{o}_{iv}$ and a confidence $c_{iv} = \phi(\pi_{iv})$, and for an ordinal variable a distribution over levels with an expected level and a confidence. A dependent variable v has a gate $g(v) \in \mathcal{B}$, and its answer is interpreted through the gating operator

$$\tilde{o}_{iv} \;=\; \begin{cases} \hat{o}_{iv} & \text{if } p_{i,g(v)} > \tau_g, \\ o_v^{\varnothing} & \text{otherwise,} \end{cases} \qquad \tau_g = 0.5, \tag{1}$$

so that a detail question is read only when its presence question has fired. The empirical justification for the operator is the leakage rate

$$\lambda_v \;=\; \Pr\!\left(\hat{o}_{iv} \neq o_v^{\varnothing} \;\middle|\; p_{i,g(v)} \leq \tau_g\right), \tag{2}$$

the share of narratives in which the model would have asserted a detail whose presence question it had itself answered in the negative. Both quantities are computed when the raw answers are flattened, and the raw answer is kept beside the gated one so that Eq. (2) can be reported for every gated variable.

The schema holds 27 questions, of which 16 are presence questions, 8 are categorical, 3 are ordinal and 7 are gated, and Table A.1 in Appendix A reproduces every instruction and criterion as sent to the model. Seven design rules govern the schema, and they were derived from the documented failure modes of the model class. Each question makes one narrow judgment, presence comes before detail with the detail gated by Eq. (1), and every categorical question carries an explicit no-match option. Named entities such as drugs are selected from a closed option set rather than generated, and numbers are pre-bucketed in the criteria text, so that a stage of pregnancy is described in weeks and months rather than left for the model to compute. No question uses a double negative or refers to another question, because questions cannot see each other, and exclusions are written into the criteria after error analysis. The effect of the last rule was measured rather than asserted. The aggression question had fired on police pursuits, which are enforcement actions rather than aggression directed at another road user, so the exclusion was added and the full schema was re-run on a fixed

sample of 2,000 narratives before and after the change. Among the 13 narratives containing pursuit language, positives fell from 4 to 1 and the mean probability from 0.395 to 0.160. The 14 other presence questions were unchanged, with label agreement of at least 0.9975 and a mean absolute probability shift of at most 0.0039, so the change was surgical rather than a general suppression. The criteria text is the entire specification of what each variable means, and the human coders and the frontier models of Section 3.3 received that same text verbatim.

Coding proceeds in two stages over a simple random sample drawn from the analyzed corpus with a fixed seed. In the first stage a lean schema of 10 presence questions is run over a random frame of 500,000 narratives, of which 499,500 returned an answer, at a cost of $25.23. The stage estimates prevalence at large sample size with exact binomial intervals and supplies a pool of likely positives for the rare classes that a purely random second-stage sample would barely contain. In the second stage the full schema is run over a frame with two parts. The first part is a random subset of 150,000 narratives from the first-stage sample, and it is the inferential backbone, because every population estimate in this paper is computed on it alone. The second part enriches each rare class from keyword hits and from first-stage positives up to a per-class ceiling, so that per-variable calibration and precision are estimable for classes that are otherwise too rare to measure. It is never pooled with the random part for a prevalence estimate. Every narrative carries its stratum and its inclusion probability, recorded at coding time because they cannot be recovered afterwards, and estimates that use the enriched part are inverse-probability weighted. The enriched allocation is trimmed proportionally within class so that the projected spend stays below its budget line, and the trim is folded into the recorded inclusion probabilities. The frame is shuffled before running, so that an interrupted run would leave a representative prefix rather than a truncation that drops the classes ordered last. The second stage coded 195,857 narratives with no errors at a cost of $30.22.

Input tokens, and therefore cost, are well described by a linear model in the number of questions k in a call and the narrative length ℓ_i in characters,

$$T(k, \ell_i) \;=\; \alpha + \beta k + b\,\ell_i, \tag{3}$$

fitted on a sweep of 7 question counts over 420 calls at a cost of $0.03. Let ϖ be the price per token and $\bar{\ell}$ the mean narrative length. A design that screens every narrative with a lean schema of k_s questions and then codes a fraction q of them with the full schema of k_f questions costs $N\varpi[T(k_s, \bar{\ell}) + q\,T(k_f, \bar{\ell})]$, against $N\varpi\,T(k_f, \bar{\ell})$ for coding everything once. Screening therefore pays only when

$$q \;<\; q^{*} \;=\; 1 - \frac{T(k_s, \bar{\ell})}{T(k_f, \bar{\ell})} \;=\; 1 - \frac{\alpha + \beta k_s + b\bar{\ell}}{\alpha + \beta k_f + b\bar{\ell}}. \tag{4}$$

Because βk dominates $b\bar{\ell}$ for schemas of the size used here, the break-even fraction depends on how many questions the screen drops and not on how much text is read. All runs use the pinned model identifier, and the identifier returned by the service is recorded on every call so that a silent version change is detectable after the fact. Requests run at a concurrency of 10 with exponential backoff over at most 5 attempts. The runner accumulates realized input tokens, projects the full-run cost after the first 2,000 calls, and aborts if that projection exceeds the ceiling it was given, so a mispriced schema cannot run away unattended. The total spend across every model call reported here was $55.79. The schema is released verbatim, the runner and the metric code are released with their unit tests, and every aggregated output behind a figure or table is released.

3.3. Reference sets

The first reference is the coded CRIS record. Narratives are joined to the person- and unit-level extract by crash identifier, and the coded fields are aggregated to crash level for all 5,601,890 crashes with the mapping of Table B.2. That mapping gives nine presence variables a coded counterpart and records two deviations from the original plan, the contributing-factor codes for wrong-way travel in place of the broader opposite-direction collision type, and the ill-driver factor as the reference for medical episodes. Coded fields are themselves incomplete, so every statistic computed against them is an agreement measure bounded by the reference's own error, and it is labeled as such throughout. The keyword baseline scored against that reference is a family of regular expressions per variable, listed in Table B.1 and tuned only on a held-out development sample.

The second reference is a human reference set drawn under a cell-filling design, described here as it was executed rather than as it was specified. Each variable's returned probability is cut into the cells [0,0.05); [0.05,0.3); [0.3,0.7); [0.7,0.95); [0.95,1], and narratives were drawn sequentially from the second-stage random stratum by a greedy rule

that repeatedly took a narrative filling the largest number of still-short cells, against a target of 30, 40, 40, 40, 30 per cell. A narrative drawn for one variable is labeled for all of its active variables, those with $p_{iv} \geq 0.05$, plus 3 randomly chosen low-probability variables as negative controls. 640 candidates were drawn, 137 were withheld by the identifier screen of Section 3.1, and the first 400 that cleared it were labeled, with the estimand fixed before any label existed.

A greedy draw followed by a screen has no inclusion probabilities that can be written down, so the weights are calibration weights in the sense of Deville and Särndal (1992) rather than inverse inclusion probabilities. The analysis population is the second-stage random stratum, and the cell separates the two routes into the sample exactly. A variable at or above 0.05 is always labeled, so its assignment probability is one, and a variable below it is labeled only as one of the 3 controls, so its assignment probability is 3 divided by that narrative's count of low-probability variables. A pair's design weight is its narrative's base weight divided by that assignment probability, scaled within its cell so that the weighted sample reproduces the population count of the cell. Appendix C gives the construction in full, including the rule that merges cells holding fewer than 5 labeled pairs, and both factors are recorded for every pair in the released audit table.

The base rate, the precision and the observed positive rate at a grid value are all population means of some quantity z_i, and each is estimated by the Hájek ratio

$$\hat{\theta} \;=\; \frac{\sum_{i \in s} w_i \, z_i}{\sum_{i \in s} w_i}, \tag{5}$$

where s is the labeled sample and w_i the calibration weight of pair i. The ratio of two weighted totals is consistent rather than unbiased, at the order Appendix D gives. The weights span a factor of 189 and give a design effect of 1.95, so the 2,416 usable judgments carry the information of about 1,265, which is what every interval below is built on.

Intervals resample narratives rather than judgments, because one narrative supplies up to 16 judgments and they are not independent. The cluster bootstrap draws narratives with replacement within the 5 draw routes the frame records, carrying all of a narrative's pairs and holding their weights fixed. With $B = 1{,}000$ replicates the interval is

$$\left[\hat{\theta}^{(\alpha/2)} - \bar{b},\ \hat{\theta}^{(1-\alpha/2)} - \bar{b}\right], \qquad \bar{b} = \frac{1}{B}\sum_{r=1}^{B} \hat{\theta}^{(r)} - \hat{\theta}, \tag{6}$$

where the percentile interval is re-centered by the bootstrap bias, because an absolute deviation such as calibration error can only be pushed upward by resampling noise, and the raw bounds are kept in the released result files. Where a binomial interval is wanted for a weighted count, the Kish effective sample size $n_{\mathrm{eff}} = (\sum w_i)^2 / \sum w_i^2$ replaces the raw count, and every interval obtained that way is approximate.

3 coders with graduate training in transportation safety labeled 2,465 narrative-variable pairs, 5,024 judgments in total, under the protocol and the fixed roster that Appendix C sets out. They were recruited to three different profiles so that agreement between them would not measure a shared domain inference. Each saw the redacted narrative and the same criteria text the model was given, under the rule to label what the narrative states rather than what probably happened, and saw nothing else, so neither the model's probability, nor the coded field, nor another coder's answer. 2,371 pairs were double-coded and 94 triple-coded as a shared calibration block, on which pairwise Cohen's kappa,

$$\kappa \;=\; \frac{p_o - p_e}{1 - p_e}, \tag{7}$$

with p_o the observed and p_e the chance agreement, runs from 0.95 to 1.00 with a median of 0.95. That coder-to-coder statistic is unweighted, because it describes the panel rather than the population, whereas every kappa comparing the model with the human reference carries the calibration weights and is labeled weighted. Raw agreement across all pairs is 99.0%, high partly because the frame is dominated by clear negatives, and the median judgment took 7.4 seconds. A pair is resolved by majority, and ties and pairs any coder marked unclear are dropped from every rate, which leaves 2,416 usable judgments with 48 disputed pairs and 18 unclear answers. Scoring the model against a label the coders could not agree on would measure nothing, so the reference is defined on the cases people find clear.

The third reference is a pair of frontier generative models from different vendors, run on the same 400 narratives and the same 2,465 pairs, and scored against the same human labels with the same weights and metric code. They received the criteria text verbatim and the same per-narrative variable assignment, and nothing else, so neither the coded fields, nor the base rates, nor the typed model's probabilities. A generative model does carry token likelihoods,

but those describe the words it chose rather than its belief about the event, and neither vendor's interface exposed them here. Each arm was therefore asked to state a probability and to use the full range, which makes the elicited number a verbalized confidence rather than a read-out of the model's internals. The arms ran through an agent harness in 20 batches of 20 narratives, with all criteria in context and no constraint on intermediate reasoning, so they had more context and more inference-time computation per judgment than the typed model. Their cost was not metered by that harness, which is why the price comparison in the results prices a comparable tier rather than these arms. Two exact statistics complete the toolkit, of which the first is the Clopper–Pearson interval that prevalences and confirmation rates carry for x successes in n trials,

$$[\,L,\,U\,], \qquad L = \begin{cases} 0 & x = 0 \\ B^{-1}\!\left(\tfrac{\alpha}{2};\, x,\, n-x+1\right) & x > 0 \end{cases}, \qquad U = \begin{cases} 1 & x = n \\ B^{-1}\!\left(1-\tfrac{\alpha}{2};\, x+1,\, n-x\right) & x < n \end{cases}, \tag{8}$$

where B^{-1} is the beta quantile function. The two boundary cases are stated because the beta quantile is undefined at them. Paired comparisons of two methods on the same narratives use the exact McNemar test on the discordant counts n_{10} and n_{01},

$$P \;=\; \min\!\left\{1,\; 2 \sum_{j \le \min(n_{10},n_{01})} \binom{n_{10}+n_{01}}{j} 2^{-(n_{10}+n_{01})}\right\}. \tag{9}$$

3.4. Evaluation

Calibration is assessed with proper scores, binned and exact calibration errors, and the logistic recalibration statistics. With weights w_i normalized to sum to the sample size, the Brier score and its Murphy decomposition over bins k with weight W_k, mean prediction $\bar{p}_k$ and observed rate $\bar{y}_k$ are

$$\mathrm{BS} = \frac{1}{n}\sum_i w_i (p_i - y_i)^2, \qquad \mathrm{BS} \approx \underbrace{\sum_k \frac{W_k}{n}(\bar{p}_k - \bar{y}_k)^2}_{\text{reliability}} - \underbrace{\sum_k \frac{W_k}{n}(\bar{y}_k - \bar{y})^2}_{\text{resolution}} + \underbrace{\bar{y}(1-\bar{y})}_{\text{uncertainty}}, \tag{10}$$

where the approximation is exact on the binned representation. The expected and maximum calibration errors over equal-mass bins are

$$\mathrm{ECE} = \sum_k \frac{W_k}{n}\left|\bar{p}_k - \bar{y}_k\right|, \qquad \mathrm{MCE} = \max_k \left|\bar{p}_k - \bar{y}_k\right|. \tag{11}$$

The model returns probabilities on a discrete grid G, so a binning-free estimator exists. Grouping by attainable value $v \in G$ with weight W_v and observed rate r_v gives

$$\mathrm{ECE}_G = \sum_{v \in G} \frac{W_v}{n}\left|v - r_v\right|, \tag{12}$$

which needs no binning parameter. It is the estimator used for the human reference set, for every bootstrap interval and for the reliability curves. The coded-field errors tabulated in the results use the equal-mass estimator of Eq. (11) computed upstream, and the two agree at the printed precision except for one variable noted there. The calibration slope and intercept are the maximum-likelihood coefficients of

$$\Pr(y_i = 1 \mid p_i) = \sigma\big(a + b\,\mathrm{logit}(p_i)\big), \tag{13}$$

with $a = 0$ and $b = 1$ under calibration. A slope above one means probabilities that are not extreme enough and a slope below one means probabilities that are too extreme, and quasi-complete separation is flagged rather than reported as a slope. The Spiegelhalter statistic

$$z = \frac{\sum_i w_i (y_i - p_i)(1 - 2p_i)}{\sqrt{\sum_i w_i^2 (1-2p_i)^2\, p_i(1-p_i)}} \tag{14}$$

is standard normal for independent observations with known probabilities. The reference set is neither weighted nor independent in that sense, so the statistic is reported here as a descriptive index and its reference distribution is obtained by the cluster bootstrap of Eq. (6) instead. Its sign is not interpreted, because the factor $1-2p$ in the summand reverses above one half, and the direction of the miscalibration is read from the weighted mean probability minus the prevalence, which Appendix D justifies.

The discrete grid has a consequence that the continuous calibration literature does not treat, and it is stated here as a proposition. Let δ be the smallest positive value in G, w_0 the probability mass the model places on exactly zero, and ρ the base rate of the variable.

Proposition 1 (Resolution floor). *If the model is calibrated then $\rho = \mathbb{E}[p] \geq \delta(1-w_0)$, so calibration requires $w_0 \geq 1-\rho/\delta$. If $\rho < \delta(1-w_0)$, no allocation of grid values is calibrated, the mean probability exceeds the prevalence, and*

$$\mathrm{ECE}_G \;\geq\; \mathbb{E}[p]-\rho \;\geq\; \delta(1-w_0)-\rho \;>\; 0. \tag{15}$$

The proof is given in Appendix D and takes five lines. The model audited here never returned zero in 2.4 million answers, so $w_0 = 0$ and the bound is $\delta - \rho$ with $\delta = 0.01$, and the proposition applies to every variable whose base rate lies below 0.01. The base rates that decide this are the coded-field prevalences of Table 4, so every application of the proposition in this paper is an application against the coded reference and is named as such. For such a variable the mean probability is not a prevalence estimate, whatever the model's ranking quality.

Selective prediction is evaluated in two forms, the conventional one and the one defined over flagged records. Let $\kappa_{iv} = \max(p_{iv}, 1-p_{iv})$ be the confidence of a presence decision and θ an abstention threshold, which is a different quantity from the decision threshold τ of Table 1 and is given its own symbol for that reason. The risk-coverage curve and its area are

$$C(\theta) = \Pr(\kappa \geq \theta), \qquad R(\theta) = \Pr(\hat{y} \neq y \mid \kappa \geq \theta), \qquad \mathrm{AURC} = \int_0^1 R\,\mathrm{d}C, \tag{16}$$

integrated as a step function over tied confidence values, with the excess area measured against an oracle that ranks every correct decision above every incorrect one. The coverage attainable at a target risk r is $C^*(r) = \max\{C(\theta) \mid R(\theta) \leq r\}$. This formulation carries little information when the positive class is rare. A rule that answers in the negative everywhere has full-coverage risk $R(0) = \rho$, so any target $r \geq \rho$ is already met by a rule that flags nothing and reads nothing. An error target at or above the base rate therefore certifies a screener that does no work, which is a statement about the target rather than about the model, and it does not follow from this that the model's own review budget is zero. All 9 variables with a coded reference have thresholded rates below five percent, running up to 3.64%, and 2 of them fall below the 1% target as well. The quantity a safety office faces is instead defined over the records the model flags. Let $F = \{i \mid p_{iv} > \tau\}$ be the flagged set, ranked by probability, and for an acceptance threshold $t \geq \tau$ define the coverage and precision

$$c(t) = \frac{\sum_{i\in F} w_i\,\mathbb{1}[p_{iv} \geq t]}{\sum_{i\in F} w_i}, \qquad \pi(t) = \frac{\sum_{i\in F} w_i\,\mathbb{1}[p_{iv} \geq t]\,y_{iv}}{\sum_{i\in F} w_i\,\mathbb{1}[p_{iv} \geq t]}, \tag{17}$$

so that $1-\pi(t)$ is the risk of accepting the top share $c(t)$ without reading it. The review budget at target precision π^* is the share of flagged records that cannot be auto-accepted,

$$H(\pi^*) \;=\; 1 - \max\big\{\, c(t) \mid \pi(t) \geq \pi^* \,\big\}, \tag{18}$$

which is the construction of Eq. (16) restricted to the flagged set with risk defined as one minus precision. Two conventions are needed to complete the definition of that maximum. If no acceptance threshold on the grid attains π^* the feasible set is empty, the maximum is taken to be zero and $H(\pi^*) = 1$, so a variable no threshold can certify routes every flagged record to review. Ties on the grid are never split, because a threshold cannot separate two records carrying the same probability, and a tied block is therefore accepted or reviewed whole.

Selecting t and certifying the precision it delivers on the same labels reports the best of many thresholds as though it were the value of one. Against the human reference the budget is therefore cross-fitted, with t chosen on one half of the narratives and both the coverage it buys and the precision it delivers measured on the other, in both directions and

over repeated splits. The in-sample value is reported beside it so that the difference is visible. The delivered precision carries a lower confidence bound from Eq. (8) at the effective accepted count, which is approximate for the reason given in Section 3.3. The budget is reported in absolute terms per year by multiplying the flagged share of the corpus by 557,564, the mean annual count of analyzed narratives, and that product counts variable-level review decisions. Where the quantity wanted is the number of documents a reader opens, the union of the flagged sets across variables is reported instead and named as such.

The two references disagree in a direction that the structure of coded-field error predicts, and the prediction is stated as a lemma. Let $r_v = \Pr(y = 1 \mid p = v)$ be the true observed rate at grid value v, and let the coded field miss a factor the narrative states with probability f_v while never recording a factor the narrative does not state, so that $r'_v = (1 - f_v)\, r_v$.

Lemma 2 (One-sided reference noise). *For every grid value, $r'_v \le r_v$, so the reliability curve measured against the coded field lies on or below the true curve. For every grid value at which $(1 - f_v)\, r_v \ge v$, $|v - r'_v| \le |v - r_v|$, and if that condition holds at every grid value carrying weight then*

$$\mathrm{ECE}_G(y') \;\le\; \mathrm{ECE}_G(y). \tag{19}$$

The proof is given in Appendix D beside the previous one. The lemma is conditional, and its condition is that the coded field never records a factor the narrative does not state. That is an assumption about this pair of sources rather than a property of coded data in general, and nothing in the literature establishes it. The human reference set supplies the empirical check, because the cell in which the coded field records a factor the model does not flag is observable and is reported in Section 4.4. The results also report the measured ratio of the two calibration errors as the companion of the bound, and that ratio turns out to be close to one, so the bound is not where the difference between the two references shows itself. Post-hoc recalibration is assessed as an add-on rather than as the headline. Platt scaling and isotonic regression are fitted on one half of the narratives and scored on the other half, over 50 repeats in both directions,

$$q^{\mathrm{Platt}}(p) = \sigma(a + b\,\mathrm{logit}(\tilde{p}))\,, \qquad \hat{m} = \arg\min_{m\uparrow} \sum_i w_i \big(m(p_i) - y_i\big)^2, \qquad q^{\mathrm{iso}}(p) = \hat{m}(p), \tag{20}$$

where $\tilde{p}$ is p clipped to $[10^{-6}, 1 - 10^{-6}]$ before the logit is taken. The clip applies in fitting and in evaluation alike, because the logit is undefined at the endpoints and the output grid reaches them. The isotonic minimization is over non-decreasing maps m and defines a fitted function, which is then evaluated at p and held constant outside the range it was fitted on. The Platt map is fitted without a penalty, so it is a two-parameter correction of the intercept and the slope, and temperature scaling is the special case with $a = 0$. Splitting on narratives rather than on judgments matters here for the reason given in Section 3.3. Maps are fitted per variable wherever the reference set carries at least 20 positive labels for that variable, and pooled otherwise, because an analyst recalibrates one variable at a time. The decision threshold τ is also retuned per variable to maximize weighted F_1 on one random half of the narratives and scored on the other, over 200 repeats, so that a threshold failure can be told from a model failure by the out-of-fold gain. Every quantity in this section is computed by the released code, and Table D.1 names the function and the result file for each equation.

4. Results

4.1. Population coding, cost, and the probability landscape

The second stage coded 195,857 of the 5,018,080 narratives in the analyzed corpus with the full 27-question schema, and 150,000 of them form the random stratum on which every population estimate rests (Table 2). Coding cost $30.22 in total, which is a realized $0.154 per thousand narratives, so coding the whole corpus at that rate would cost $774. The median latency was 0.20 seconds, the ninetieth percentile 0.33 seconds, the sustained rate about 44 requests per second, and the mean input 3,673 tokens per narrative, with 0 failed calls. The fitted token model of Eq. (3) has $\alpha = 171$, $\beta = 122$ and $b = 0.236$ tokens per character, with $R^2 = 0.987$ for the question-count model (Figure 2d), and Table 2 also reports the direct fit at 27 questions, whose single intercept absorbs the schema term. Its shape is the substantive finding of the run, because at 27 questions the schema accounts for about 3,451 tokens per call while a median narrative of 378 characters contributes about 89. Each additional question costs a constant 122 tokens regardless of the corpus, so a lean screen is cheaper only in proportion to the questions it omits. Two break-even fractions follow and they are not the same quantity. Evaluating Eq. (4) with the fitted token model at the

median narrative length, with 10 questions in the screen against 27 in the full schema, predicts that screening pays only when fewer than a fraction 0.58 of narratives go on to full coding. The screen as executed cost 0.33 of the full call per narrative, which puts the realized break-even at 0.67. The two differ because the lean schema carries shortened instruction text as well as fewer questions, so its intercept is below the one the question-count model assumes, and the executed screen is therefore cheaper than the model predicts. Under either figure the conclusion is the same, and asking more questions in one call is the dominant efficiency lever.

Table 2
Corpus and run statistics.

Quantity	Value
Crash records in the narrative file	5,109,746
Narratives empty	91,175 (1.78%)
Narratives analyzed (>40 chars)	5,018,080
Narrative length, median / p95 / p99 (chars)	378 / 935 / 1427
Stage-2 narratives coded	195,857
of which random stratum	150,000
Model	jev-1.13.0 (195,857)
Mean input tokens per narrative	3,673
Median latency (s)	0.20
Token model, direct fit at 27 questions	$a = 3562$, $b = 0.236$/char, $R^2 = 0.975$
Schema overhead per question (tokens)	122 ($R^2 = 0.987$)
Cost per 1,000 narratives, realised (USD)	0.1543
Stage-2 spend (USD)	30.22

Note: Prices are TypeSafe list prices accessed 2026-09-17 and are quoted rather than endorsed.

The returned probabilities pile up at the two ends of the grid in every one of the 16 presence variables (Figure 3). Most narratives receive a value at or near the floor, and every panel turns up again at the ceiling, where the top decile holds more mass than the decile below it in all 16 of 16 variables. The uncertain band between 0.3 and 0.7 holds under three percent of narratives for every variable. The two ends do not carry equal weight everywhere. For 6 variables, namely medical episode, aggression, intentional, wrong way, unbelted, witness cited, the middle of the range holds more narratives than the ceiling does, so the upper mode is a rise rather than a second peak. That shape is favorable for selective prediction, because the population a threshold would route to review is small. The model returns its probabilities on a discrete grid, and this property was not documented by the developer. The returned values take at most 100 distinct levels, running from 0.01 to 1.00 in steps of 0.01, and across 2.4 million answers on the random stratum the value zero occurred 0 times and the value one occurred 2 times. The same grid governs the confidence attached to categorical and ordinal answers. The lower end of the grid has a first visible consequence in Table 3. The expected probability is the estimate a calibrated model would match to the true prevalence, but for the rarest variables most narratives sit at the grid's smallest value, so it largely measures the floor rather than the data. For preg mentioned, 80% of the expected probability is contributed by narratives at $p = 0.01$. 5 variables exceed the 20% reporting threshold, one of them by a margin that rounds away in print, and they are marked in the table, where thresholded rates or keyword rates are the appropriate summaries. Gating behaves as designed, because the measured leakage rate of Eq. (2) across the 7 gated variables ranges from 0.03% to 3.99%, with the maximum for medical type, where 7,577 of 189,733 gate-off narratives carried a substantive answer that the operator suppressed.

4.2. Calibration against coded fields and against human judgment

Table 4 reports calibration against the coded CRIS fields for the 9 variables that have a usable reference, and Figure 4 draws the reliability curves on the model's own output grid with the human reference set in the same panels. Against the coded reference the tabulated calibration error ranges from 0.012 for unbelted to 0.045 for wrong way, and the calibration slope of Eq. (13) from 0.51 to 1.24. Every calibration error reported against either reference in this paper is the exact grid error of Eq. (12) rather than the binned error of Eq. (11). Each interval is a bootstrap of that same quantity, so every point estimate lies inside its own interval and the two references can be compared with each other. Slopes on both sides of one appear, so the departures are variable-specific rather than a single systematic bias. The curves are drawn by grid value rather than in equal-mass bins, because for a rare variable almost all mass lands

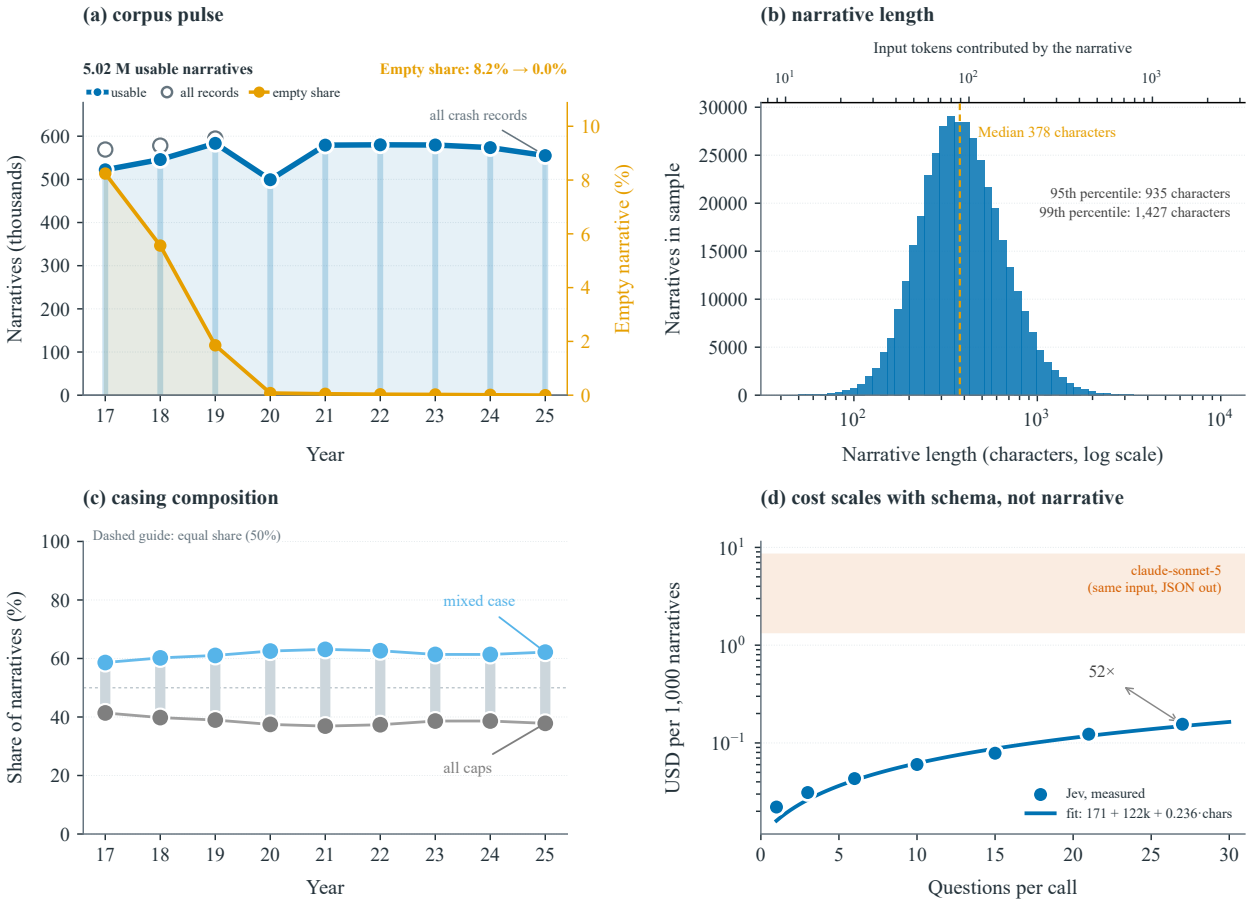


Figure 2: Corpus and cost. (a) Usable narratives per year in thousands (blue) against all crash records (hollow), with the share of records whose narrative is empty (orange, right axis) falling from 8.24% in 2017 to zero by the middle of the period. (b) Narrative length on a logarithmic axis, median and upper percentiles marked, with the input tokens on the upper axis. (c) Share of narratives in mixed case (light blue) and in capitals (gray). (d) Measured cost per thousand narratives against questions per call (blue), the fitted model of Eq. (3), and the list-price band of a frontier generative model on the same input (orange).

on two or three grid values and a ten-bin scheme collapses to a handful of populated bins. The grid floor bounds what calibration is achievable. Coded base rates run from 0.43% to 3.63%, and for 4 variables, namely drug involved, phone use, wrong way, medical episode, the base rate falls below the smallest expressible probability of 0.01. Proposition 1 therefore applies to them, and the bound of Eq. (15) on the attainable calibration error runs from 0.0046 to 0.0057, with the maximum for phone use. The floor and the coded base rate are marked on every panel of Figure 4 so that the affected cases are visible rather than inferred.

Against the human labels the calibration error is almost unchanged, and the difference between the two references appears elsewhere. Table 5 scores the model over 16 variables and 2,416 usable judgments. Per variable the calibration error against the human labels is 0.8 to 2.2 times the error against the coded field, with a median ratio of 1.0, and it is the larger of the two for 7 of 9 variables. The margin does not survive rounding to the three decimals Figure 4 prints for 2 of them, which is why a count taken from the figure's annotations comes out lower. Once both references are weighted to the same population and measured with the same estimator, neither one makes the model's probabilities look materially better calibrated than the other, and Figure 4 annotates the two errors side by side in each panel to show it.

The two references separate sharply on agreement rather than on calibration, and that separation is the result the rest of this paper rests on. Cohen's kappa against the coded field is below the calibration-weighted kappa against human judgment for all 9 of 9 variables that carry a coded counterpart, by a median of 0.26. The widest case is drug involved, which reaches 0.48 against the coded field and 0.96 against human judgment, so the same extraction reads

Table 3
Prevalence estimates on the Stage-2 random stratum.

variable	keyword % of 5,018,080	$p > 0.5$ % of 150,000	E[p] % of 150,000	floor %	hits	confirmed % of hits	uncertain % of 150,000
hit_and_run	–	15.863 [15.679, 16.049]	17.384	0	–	–	2.02
witness_cited	–	6.127 [6.006, 6.249]	9.033	0	–	–	2.87
transported	–	4.997 [4.887, 5.108]	6.778	2	–	–	0.42
alcohol_involved	–	3.635 [3.541, 3.731]	5.498	4	–	–	0.38
hydroplane	0.895	2.352 [2.276, 2.430]	3.623‡	20	1,328	99.77 [99.34, 99.95]	0.41
vehicle_defect	0.500	2.209 [2.135, 2.284]	3.965	8	728	93.13 [91.05, 94.86]	0.91
animal_involved	1.997	2.016 [1.945, 2.088]	3.224‡	25	2,981	83.76 [82.39, 85.07]	0.16
wrong_way	0.332	1.427 [1.368, 1.489]	5.087	0	513	78.36 [74.54, 81.85]	1.85
fatigue	1.170	1.399 [1.340, 1.459]	2.653‡	28	1,824	96.77 [95.85, 97.53]	0.15
aggression	0.566	1.103 [1.050, 1.157]	3.916	4	842	48.46 [45.03, 51.89]	1.11
medical_episode	0.742	1.055 [1.004, 1.108]	2.984	15	1,148	82.06 [79.71, 84.23]	0.77
phone_use	–	1.038 [0.987, 1.091]	2.463‡	27	–	–	0.28
unbelted	–	0.596 [0.558, 0.636]	2.844	4	–	–	0.79
drug_involved	0.242	0.562 [0.525, 0.601]	2.501	12	377	88.06 [84.36, 91.16]	0.26
intentional	0.165	0.359 [0.329, 0.390]	3.267	4	225	59.56 [52.83, 66.03]	0.54
preg_mentioned	0.108	0.116 [0.099, 0.135]	1.179‡	80	173	97.11 [93.38, 99.06]	0.01

Note: Keyword % is over all 5,018,080 narratives. Jev % thresholds at $p > 0.5$ with Clopper–Pearson intervals. E[p] is the mean probability. The floor column is the share of E[p] from narratives at the smallest probability the model can express, $p = 0.01$; the double dagger marks a share above 20%, where E[p] is inflated by the output grid and is not a prevalence estimate (Section 4.1). Confirmation % is the share of keyword hits Jev also calls positive; uncertain % the share with $p \in [0.3, 0.7]$.

as mediocre under one reference and as strong under the other. The narrowest agreement of all is unbelted at 0.10 against the coded field, and there the human reference agrees that the model is weak, at 0.45. The mechanism is the one Lemma 2 describes, namely one-sided omission. A factor the narrative states and the coded field does not record turns a correct extraction into an apparent false positive. That costs the model agreement without costing it calibration, because its probabilities stay close to the rate at which the factor is actually stated. The practical consequence is that a coded reference understates how faithfully the model reads a narrative, and Section 4.5 prices what those unrecorded factors are worth.

Calibration against the human reference is nonetheless rejected, and the failure is a level shift rather than over-confidence. Pooled across variables the calibration error is 0.0231 with a cluster-bootstrap interval of [0.0212, 0.0245], the Brier score is 0.0045 with a skill score of 0.828, and the Spiegelhalter statistic of Eq. (14) is -5.3 with a bootstrap standard error of 0.23. The calibration slope is 1.63, above one, so the probabilities are not extreme enough and the model hedges toward the middle of the range, which is the opposite of the over-confidence usually reported for neural classifiers. The probabilities also sit too high overall, with a mean of 4.8% against a weighted base rate of 2.7% and an intercept of -0.82, an over-statement of about 80% in relative terms. A safety office summing probabilities to estimate prevalence would therefore over-count, whereas one ranking narratives for review is unaffected, because ranking is invariant to a monotone transformation, and that invariance is why the remedy is a recalibration rather than a retrained

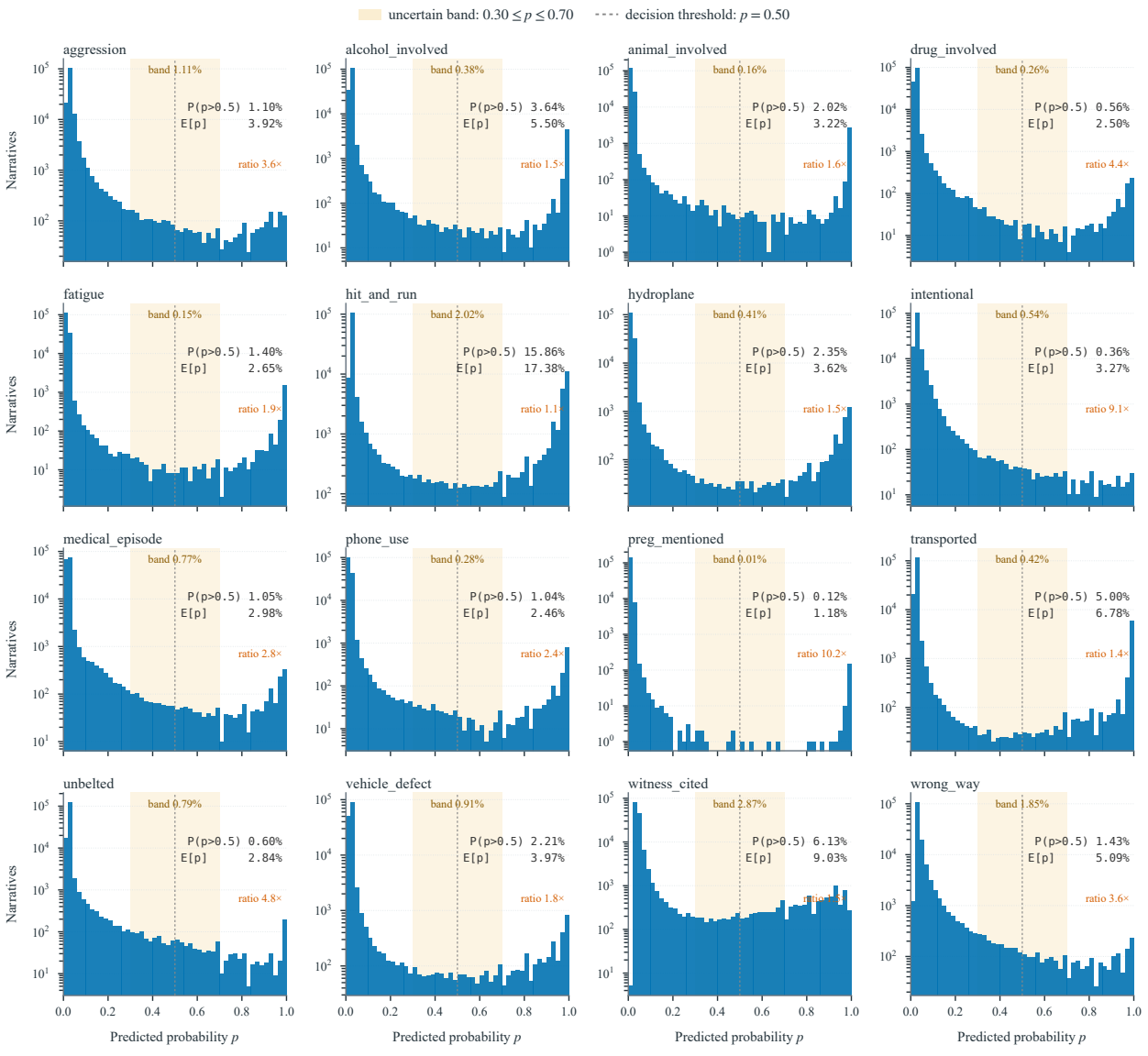


Figure 3: Returned probabilities for each presence variable on the random stratum, with a logarithmic count axis. The shaded band is the uncertain region from 0.3 to 0.7 with its share annotated and the dashed line the decision threshold, and each panel gives the thresholded rate, the expected probability and their ratio. Every panel spikes at the floor and turns up again at the ceiling.

model. Figure C.1 in Appendix C draws the same human-referenced reliability for the seven variables without a coded counterpart.

Recalibration recovers most of the error at a small cost in labels. Table 6 fits the maps of Eq. (20) on half of the narratives and scores them on the other half in both directions, over 50 repeats. Splitting on narratives rather than on pairs matters here, because one narrative supplies up to sixteen judgments and a split over pairs would fit the map partly on the records it is then scored against. Out of fold, the pooled calibration error falls from 0.0231 to 0.0069 under Platt scaling and to 0.0069 under isotonic regression, a factor of 3.3, with an interval of [0.0045, 0.0112] on the Platt value. The fitted map is a two-parameter correction with an unpenalized intercept and slope, and after it the pooled slope is 0.97 against 1.63 before, so the map removes the level shift rather than merely rescaling it.

The pooled map is a demonstration and the per-variable maps are the deployment evidence, since an analyst recalibrates one variable at a time. Table 6 therefore carries a per-variable block for the 11 variables whose reference labels contain at least 20 positives. The held-out calibration error under a variable's own Platt map runs from 0.0007

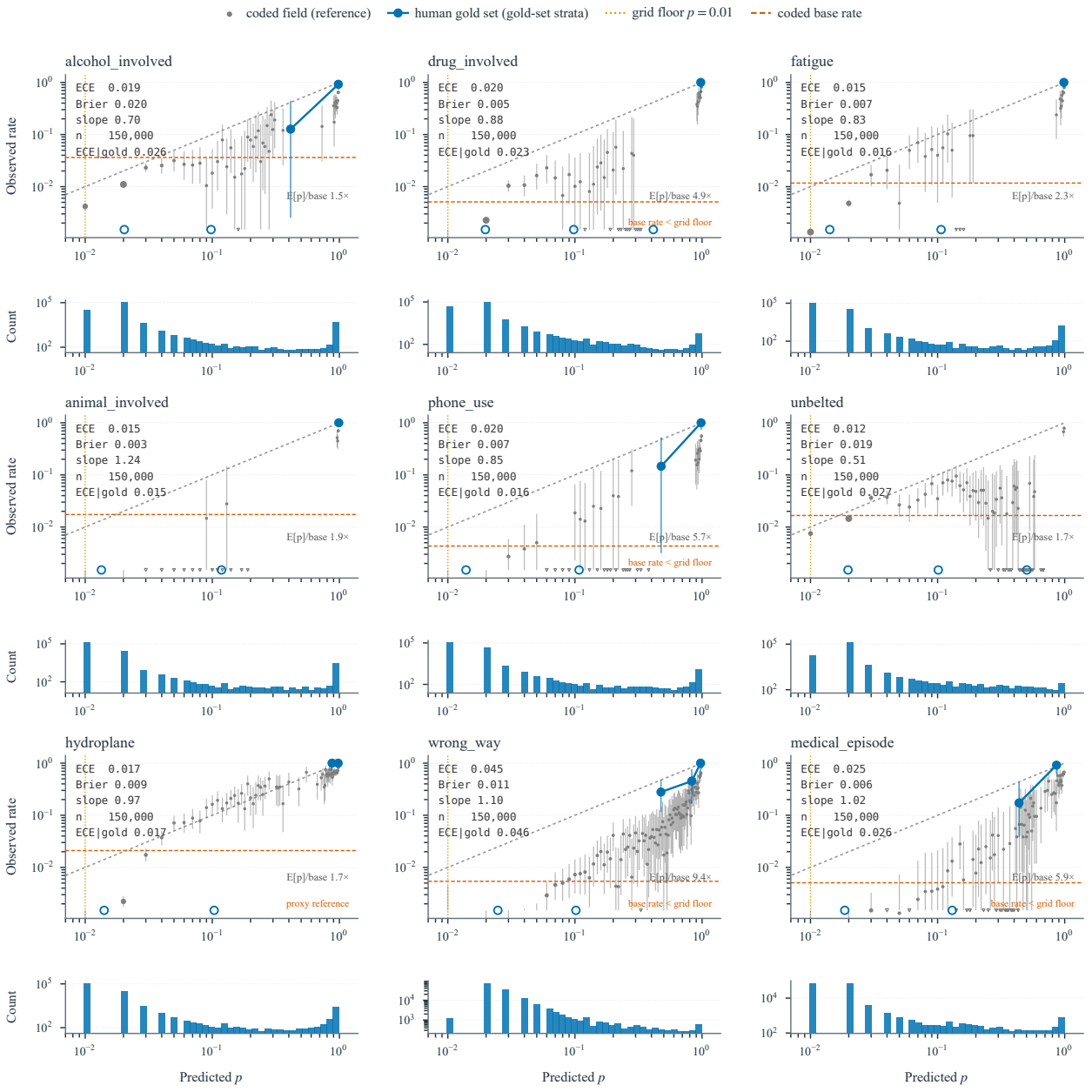


Figure 4: Reliability against two references for the nine variables with a coded counterpart. Gray markers are the observed rate in the coded CRIS field at each attainable grid value, with exact binomial intervals and marker area proportional to support; observed rates of zero are marked on the floor. Blue markers are the human reference set on its calibration cells, weighted by the design weights of Section 3.3, with approximate intervals at Kish's effective sample size. Open circles are strata with no positive, and strata too thin to support a rate are not drawn, which is why the restraint-use panel carries no filled blue marker. Axes are logarithmic. The dotted vertical guide is the grid floor and the dashed horizontal guide the coded base rate, and where the latter lies below the former no allocation of grid values can be calibrated. Each panel annotates the calibration error against both references.

Table 4
Calibration against coded CRIS fields (agreement).

			Error					Cox calibration		
variable	**coded field**	*n*	**ECE**	**MCE**	**Brier**	**rel.**	**res.**	**slope**	**intcpt**	*z*
alcohol_involved	coded_alcohol	150,000	0.0187 [0.0180, 0.0194]	0.022	0.0197	0.0004	0.0003	0.70	-1.81	-1.5
drug_involved	coded_drug	150,000	0.0199 [0.0195, 0.0203]	0.025	0.0054	0.0004	0.0000	0.88	-2.57	-42.5
fatigue	coded_fatigue	150,000	0.0148 [0.0144, 0.0152]	0.034	0.0069	0.0003	0.0003	0.83	-2.42	-23.0
animal_involved	coded_animal	150,000	0.0149 [0.0146, 0.0152]	0.035	0.0031	0.0003	0.0011	1.24	-3.28	-34.9
phone_use	coded_phone	150,000	0.0203 [0.0199, 0.0207]	0.043	0.0070	0.0007	0.0000	0.85	-3.58	-27.2
unbelted	coded_unbelted	150,000	0.0122 [0.0115, 0.0128]	0.013	0.0186	0.0002	0.0000	0.51	-2.21	-13.1
hydroplane†	coded_hydro_proxy	150,000	0.0166 [0.0162, 0.0171]	0.030	0.0090	0.0003	0.0012	0.97	-1.59	-22.3
wrong_way	coded_wrongway	150,000	0.0455 [0.0450, 0.0460]	0.188	0.0112	0.0048	0.0002	1.10	-3.09	-64.0
medical_episode	coded_medical	150,000	0.0248 [0.0244, 0.0252]	0.037	0.0065	0.0008	0.0000	1.02	-2.99	-43.3

Note: Coded CRIS fields are themselves incomplete, so these measure *agreement* rather than accuracy; accuracy against the human reference set of Section 3.3 is in Table 5. Incompleteness costs agreement rather than calibration, because a factor the narrative states and the code omits turns a correct extraction into an apparent false positive. The dagger marks a coded field that is a proxy rather than a direct match. ECE and its interval are both the exact estimator on the model's two-decimal output grid, so every point estimate lies inside its own interval. The interval is a bootstrap over crashes (B = 1,000), the sampling unit of the stratum.

to 0.0339 with a median of 0.0034, against a median of 0.0261 for the raw output, and 10 of the 11 variables improve under it. The remaining 5 variables carry too few positives to estimate their own map and can only be recalibrated under the pooled one, which is a statement about the size of the reference set rather than about the method. The practical claim is that a few thousand human judgments, about 3.4 hours per coder at the median response time, are enough to recalibrate the variables that carry enough positive labels, over a corpus of 5,018,080 narratives.

4.3. Accuracy, selective prediction, and the review budget

Against human judgment of the same criteria text, and calibration-weighted to the population under the design of Section 3.3, pooled precision is 0.902 with an interval of [0.877, 0.926] and recall is 0.915. Pooled F_1 is 0.908 with an interval of [0.871, 0.936], the weighted Cohen's kappa is 0.906, and 11 of the 16 variables reach an F_1 of 0.85 or more (Table 5). Every interval here resamples narratives rather than judgments, which is the unit the reference set was drawn in. This subsection carries the paper's only accuracy claim, and every other number in this section is agreement with coded fields. These rates are also conditional on the pairs the coders found clear, and the bounds that conditioning puts on them are reported at the end of this subsection.

4 variables fall below an F_1 of 0.70, namely unbelted, intentional, wrong way, vehicle defect, and they do not all fail the same way. For 3 of them, unbelted, intentional, wrong way, the failure is precision, and unbelted reaches only 0.29 at full recall. For 1 of them, vehicle defect, the failure is the opposite, with precision 0.96 and recall 0.54, so the model states the factor accurately when it states it and misses more than a third of the records that carry it. Only 5 of the 48 disputed pairs fall in these variables, so the coders were essentially unanimous and these are model errors rather than ambiguous criteria, which is why Table 5 is presented per variable. It would be natural to conclude that these concepts are simply not recoverable from a crash narrative, but Section 4.4 shows that conclusion to be wrong for most of them. Two frontier models given the same narratives and the same criteria recover 3 of the 4 weak variables above an F_1 of 0.80, namely unbelted, intentional, vehicle defect, reaching 0.91 on intentional acts, 0.88 on restraint use and 0.99 on vehicle defects. The information is therefore in the narrative for those, and the failures belong to the

Table 5
Accuracy against the human reference set.

			Accuracy vs. human labels					Calibration				
variable	n	base %	prec.	rec.	F_1	κ weighted	κ vs. coded	ECE	vs. coded	gap	slope	$H(0.90)$ % held out
preg_mentioned	107	0.2	1.00	1.00	1.00	1.00	–	0.011	–	–	sep.	5
animal_involved	129	1.9	1.00	0.98	0.99	0.99	0.91	0.015	0.015	1.0×	sep.	0
hit_and_run	205	16.1	0.99	0.96	0.97	0.97	–	0.035	–	–	2.12	0
transported	146	4.4	0.94	1.00	0.97	0.97	–	0.024	–	–	sep.	0
fatigue	120	1.3	0.93	1.00	0.97	0.97	0.71	0.016	0.015	1.1×	sep.	0
hydroplane	130	2.1	0.96	0.95	0.96	0.96	0.75	0.017	0.017	1.0×	2.35	0
drug_involved	148	0.4	0.92	1.00	0.96	0.96	0.48	0.023	0.020	1.1×	sep.	0
phone_use	123	0.8	0.93	0.97	0.95	0.95	0.48	0.016	0.020	0.8×	sep.	0
alcohol_involved	140	3.0	0.88	1.00	0.94	0.93	0.70	0.026	0.019	1.4×	sep.	5
aggression	165	0.9	0.86	0.91	0.88	0.88	–	0.029	–	–	3.23	11
witness_cited	206	5.3	0.84	0.91	0.87	0.87	–	0.055	–	–	2.15	19
medical_episode	164	0.9	0.78	0.91	0.84	0.84	0.53	0.026	0.025	1.0×	sep.	14
vehicle_defect	132	3.9	0.96	0.54	0.69	0.68	–	0.032	–	–	1.39	0
wrong_way	183	1.0	0.42	0.68	0.52	0.52	0.37	0.046	0.045	1.0×	1.55	72
intentional	173	0.2	0.33	1.00	0.49	0.49	–	0.035	–	–	sep.	68
unbelted	145	0.2	0.29	1.00	0.45	0.45	0.10	0.027	0.012	2.2×	sep.	64
pooled	2,416	2.7	0.90	0.91	0.91	0.91	–	0.023	–	–	1.63	1

Note: These are accuracy rather than agreement measures, because the reference is 2,416 human judgments of the criteria text the model was given. Every rate is calibration-weighted to the Stage-2 random stratum under the design of Section 3.3, so the base rate is the population rate and both kappa columns are population quantities. Bars scale F_1 from 0.40 and red marks $F_1 < 0.70$. The gap column is ECE divided by the coded-field ECE of Table 4. The budget column is cross-fitted, the threshold chosen on one half of the narratives and certified on the other. "sep." denotes quasi-complete separation.

Table 6
Post-hoc recalibration on the human reference set, validated across narrative splits.

scope	mapping	n	ECE, held out	Brier	slope	intercept
pooled	none (raw model output)	2,416	0.0231 [0.0212, 0.0244]	0.0045	1.63	-0.82
pooled	Platt (logistic on the logit)	2,416	0.0069 [0.0045, 0.0112]	0.0036	0.97	-0.06
pooled	isotonic	2,416	0.0069 [0.0045, 0.0111]	0.0036	0.77	-0.38
animal_involved	raw / Platt	129	0.0146 / 0.0007	0.0006 / 0.0005	–	0.10
fatigue	raw / Platt	120	0.0159 / 0.0008	0.0008 / 0.0004	–	0.00
transported	raw / Platt	146	0.0240 / 0.0014	0.0020 / 0.0008	–	-0.31
hydroplane	raw / Platt	130	0.0169 / 0.0020	0.0013 / 0.0015	0.79	-0.41
medical_episode	raw / Platt	164	0.0260 / 0.0025	0.0028 / 0.0015	0.92	-0.11
aggression	raw / Platt	165	0.0290 / 0.0034	0.0037 / 0.0018	0.93	-0.06
alcohol_involved	raw / Platt	140	0.0261 / 0.0053	0.0046 / 0.0030	0.90	0.08
wrong_way	raw / Platt	183	0.0462 / 0.0103	0.0096 / 0.0059	0.96	-0.11
hit_and_run	raw / Platt	205	0.0349 / 0.0114	0.0074 / 0.0059	0.93	-0.05
witness_cited	raw / Platt	206	0.0549 / 0.0194	0.0119 / 0.0108	0.96	-0.02
vehicle_defect	raw / Platt	132	0.0324 / 0.0339	0.0173 / 0.0185	0.62	-1.20

Note: Each mapping is fitted on one half of the narratives and scored on the other, both ways and over 50 repeats, because an in-sample fit would be circular and a split over pairs would place one narrative on both sides of it. The row "none" uses the same protocol, so it is comparable to the rows below but not to the full-sample ECE of Table 5. The Platt map is unpenalised, fitting an intercept and a slope. The pooled block is one map over all variables; the per-variable block is what an analyst deploys, and variables with fewer than 20 positive labels are omitted from it.

interaction between this model and this schema rather than to the narrative. The exception is wrong way, where the best frontier arm reaches 0.77 and no arm clears the threshold, so that concept may genuinely be hard to state from a narrative alone.

The obvious remedy does not work, and the test is reported rather than the intuition. Precision 0.29 at full recall looks like a threshold set too low, particularly given the upward level shift established above, so Table 7 retunes τ per variable to maximize F_1 on one random half of the narratives and scores it on the other over 200 repeats. No variable gains as much as 0.05 in F_1 among the 10 for which the procedure is stable, the median change is -0.01, and the best is 0.03 for medical episode. The weak variables divide by how much evidence they carry. 2 of them, intentional, unbelted, appear to gain as much as 0.42, but each carries only 5 positive labels and both exceed the stability bound. The apparent improvement is a threshold fitted to noise, and the interquartile range of the chosen threshold in Table 7 shows how little the two halves agreed. The other 2, vehicle defect, wrong way, carry up to 32 positives and do not gain at all, the best of them changing by -0.01. Wrong-way travel is the clearest case, since at a tuned threshold of 0.60 on 21 positives its out-of-fold F_1 is 0.48, below what the default already achieves. The default threshold of 0.5 is therefore kept throughout, since the well-behaved variables are not improved by retuning either.

These failures belong to the pairing of this model with this schema rather than to the schema alone. The frontier arms of Section 4.4 read the same criteria text and recover 3 of the 4 weak variables, so the wording cannot be the whole explanation, and a rewrite of the criteria would repair the interaction rather than a defect in the concepts. That rewrite is not attempted here, because the reference set was drawn against the current schema and a schema tuned until its weak variables pass on the same labels would be circular.

Table 7
Per-variable decision threshold, retuned on the human reference set and validated out of fold.

			Retuned threshold, out of fold					
variable	n^+	F_1 **at** $\tau = 0.5$	**tuned** τ^*	**half-sample** τ **median [IQR]**	F_1 **tuned, out of fold**	**s.d.**	**change**	**stable?**
unbelted	5	0.45	0.77	0.75 [0.73, 0.77]	0.87	0.17	+0.42	no
intentional	5	0.49	0.66	0.66 [0.64, 0.70]	0.65	0.22	+0.16	no
wrong_way	21	0.52	0.60	0.60 [0.40, 0.76]	0.48	0.09	-0.05	yes
vehicle_defect	32	0.69	0.40	0.37 [0.15, 0.47]	0.68	0.15	-0.01	no
medical_episode	24	0.84	0.73	0.70 [0.53, 0.73]	0.87	0.07	+0.03	yes
witness_cited	41	0.87	0.42	0.42 [0.39, 0.50]	0.84	0.05	-0.04	yes
aggression	31	0.88	0.49	0.49 [0.48, 0.61]	0.85	0.06	-0.03	yes
alcohol_involved	25	0.94	0.94	0.79 [0.70, 0.94]	0.93	0.04	-0.00	yes
phone_use	15	0.95	0.66	0.59 [0.54, 0.66]	0.95	0.05	-0.00	no
drug_involved	8	0.96	0.60	0.53 [0.46, 0.60]	0.93	0.09	-0.02	no
hydroplane	31	0.96	0.54	0.27 [0.24, 0.54]	0.94	0.03	-0.02	yes
fatigue	27	0.97	0.62	0.62 [0.52, 0.66]	0.96	0.03	-0.00	yes
transported	35	0.97	0.71	0.71 [0.62, 0.71]	0.98	0.02	+0.01	yes
hit_and_run	92	0.97	0.30	0.35 [0.30, 0.43]	0.97	0.01	-0.00	yes
animal_involved	23	0.99	0.39	0.34 [0.28, 0.39]	0.98	0.04	-0.01	yes
preg_mentioned	4	1.00	0.06	0.06 [0.05, 0.06]	0.94	0.07	-0.06	no

Note: n^+ counts positive reference labels. τ^* maximizes weighted F_1 on the full reference set; the out-of-fold column picks τ on a random half and scores it on the other, both ways over 200 repeats, splitting on narratives rather than pairs, and is what retuning actually buys. The median and interquartile range are of the thresholds those half samples chose. An unstable threshold has a spread above 0.10 or fewer than 20 positives, is fitted to noise, and must not be transported.

The conventional selective-prediction formulation returns nothing on these variables. At full coverage the accuracy-based selective risk of Eq. (16) against the coded reference runs from 0.33% to 2.08% (Figure 5a). Every variable therefore sits below the 5% target usually adopted in the selective-classification literature, 5 of 9 sit below the 1% target as well, and the implied review budget is zero for 9 of 9 variables at the 5% target. Applied unchanged, the standard analysis would certify a rare-event screener as needing no human review at all. The budget over flagged records carries the information instead. Precision among flagged positives against the coded reference ranges from 0.21 for unbelted to 0.85 for animal involved, so the review effort lies entirely in the flagged set. Ranking that set

by probability and auto-accepting the most confident traces the curves of Figure 5b. Those curves are drawn as steps because the output grid is discrete, so the only operating points that exist are the attainable thresholds and the intervals between them carry the precision of the lower one. The alcohol curve shows why that distinction matters in practice. It reaches full precision only on a negligible share of its flags, and it then drops straight through the policy band to 84% as soon as the block of records at the highest grid value is accepted, which is 63% of its flags at once. Eq. (18) reads those attainable points and converts them into the budgets of Table 8. Against the coded reference the budget saturates, because 8 of the 9 variables measurable against both references require reviewing essentially every flagged record to reach 90% precision (Figure 5c), and only animal involved can auto-accept most of its flags. As annual workload for Texas, that precision across all 9 variables implies 68,049 variable-level review decisions per year, of which 20,262 are for alcohol involved alone (Figure 5d). That total counts decisions rather than documents, because a narrative that more than one variable routes to review contributes once for each of them. A reader opens such a narrative once and answers every question it raises, so the staffing figure is the union of the flagged sets, which is 61,994 narratives per year, or 11.1% of the corpus.

These budgets answer how much review is needed to match the coded field rather than how much is needed to be right, and the human reference shows the difference to be the dominant term. The budget also has to be certified on labels that did not choose it, because the acceptance threshold is selected from a grid and the best of many thresholds is not the value of one. Table 8 therefore reports the budget twice, chosen on one half of the narratives and measured on the other over 100 splits, and chosen and measured on the same half. Pooled, the held-out budget of Eq. (18) is 0.8% of flagged records at 90% precision against 0.0% in sample, so the optimism of in-sample selection is 0.8 percentage points at this pooled level. The precision that budget actually delivers out of fold is 0.906, with a lower confidence bound of 0.844 computed at the effective accepted count and therefore approximate.

The pooled budget is a mixture over variables rather than a policy, and Table 5 gives the figure an agency would act on. Held out, the per-variable budgets run from 0.0% to 72.2%, the largest being wrong way, and 6 of 16 variables already meet the 90% target with no review at all. The human-referenced budget is below the coded-field budget for all 9 comparable variables, so the coded-field budgets of Table 8 are an upper bound on human effort. The curve of Figure C.2 in Appendix C pools the reference set across variables and is read as a summary of that mixture rather than as an operating point.

One qualification applies to every accuracy and budget figure in this subsection. The 49 pairs the coders left unclear or disagreed on are excluded from all of them, so each is conditional on the pairs the coders found clear. Treating every excluded pair as a model error lowers pooled F_1 to 0.875 and raises the pooled budget to 6.9%, treating them all as model-correct raises F_1 to 0.912, and routing them to review regardless of probability raises the budget to 3.7%. Those are the bounds, and Table C.1 in Appendix C reports the exclusions by variable and by probability bin.

Table 8
Selective prediction and the human-review budget (agreement).

variable	AURC	excess	prec.	flagged %	$H(0.90)$ %	$H(0.95)$ %	human $H(0.90)$ % held out	human $H(0.90)$ % in sample	review decisions/yr at 90% precision
alcohol_involved	0.0163	0.0159	0.712	3.635	100.0	100.0	5.1	3.9	20,262
drug_involved	0.0018	0.0018	0.464	0.562	100.0	100.0	0.0	0.0	3,134
fatigue	0.0036	0.0035	0.653	1.399	100.0	100.0	0.1	0.0	7,798
animal_involved	0.0014	0.0014	0.848	2.016	7.0	100.0	0.0	0.0	792
phone_use	0.0025	0.0024	0.340	1.038	100.0	100.0	0.2	0.0	5,788
unbelted	0.0136	0.0132	0.209	0.596	100.0	100.0	63.6	71.0	3,323
hydroplane	0.0012	0.0011	0.719	2.352	100.0	100.0	0.0	0.0	13,114
wrong_way	0.0016	0.0014	0.259	1.427	100.0	100.0	72.2	70.9	7,958
medical_episode	0.0014	0.0013	0.398	1.055	100.0	100.0	14.4	18.9	5,880

Note: Agreement against the incomplete coded fields rather than accuracy; see the note to Table 4. $H(\pi)$ is the share of *flagged positives* a human must review to reach precision π; accuracy-based budgets are degenerate because the positive class is rare (Section 4.3). The two human columns are measured against the human reference set. The first picks the threshold on one half of the narratives and certifies it on the other, and the second does both on the same half, so their difference is the optimism of in-sample selection. The last column counts *variable-level review decisions* per year rather than narratives.

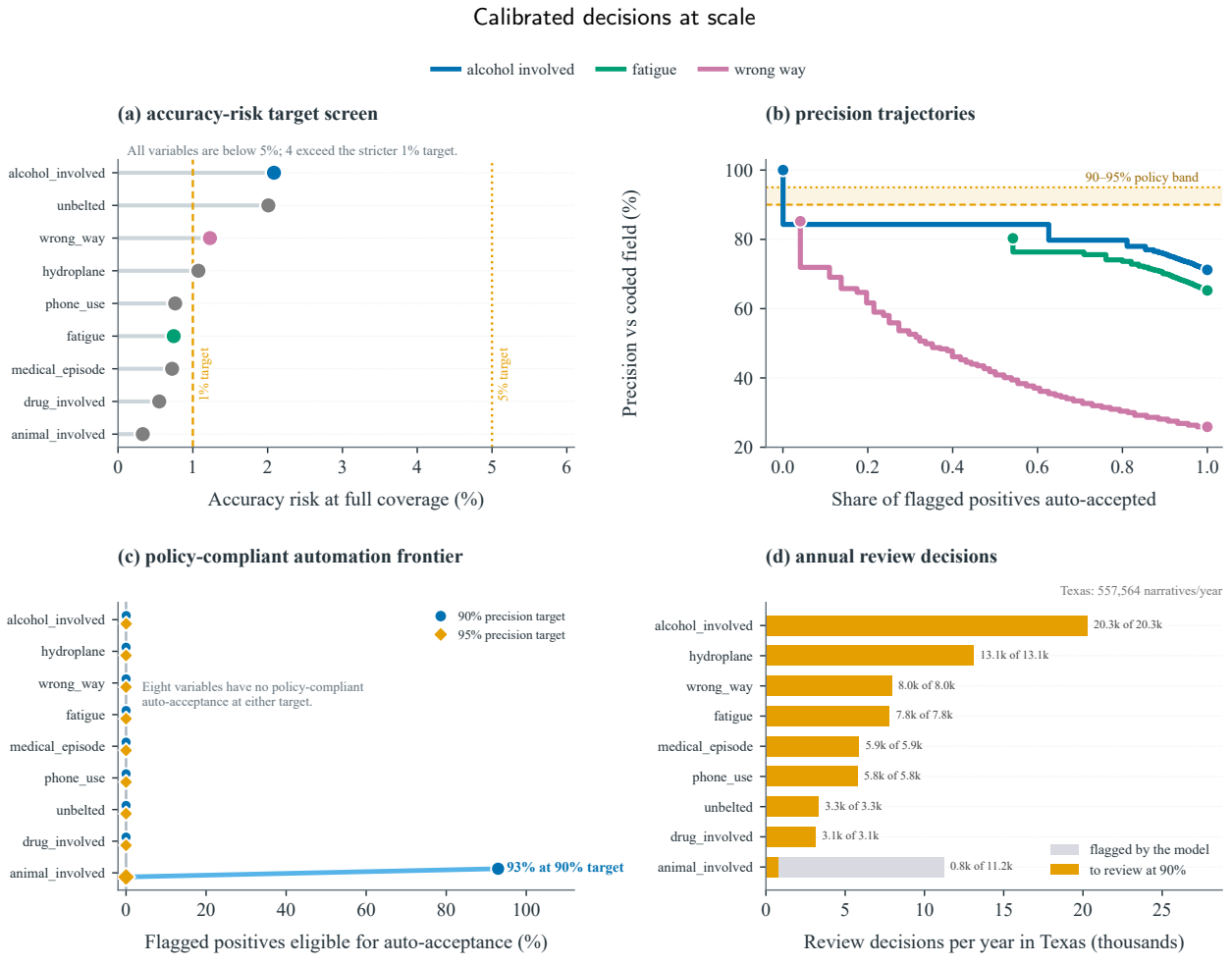


Figure 5: Selective prediction against the coded reference. (a) Accuracy-based selective risk at full coverage for each variable against the two conventional targets, with every variable below the 5% line and four above the 1% line. (b) Precision against the share of flagged positives auto-accepted for alcohol involvement (blue), fatigue (green) and wrong-way travel (pink), with the 90 to 95% policy band shaded. The curves are steps, each interval taking the precision of the lower attainable threshold, because any coverage inside it means accepting a block of records tied at one grid value. No curve rests inside the band. (c) The share of flagged positives a variable could auto-accept while holding each target, the complement of the budget of Eq. (18), zero for eight of nine variables at the lower target and for all nine at the higher. (d) The variable-level review decisions this implies for Texas each year (orange) against the volume the model flags (gray). Precision here inherits the omissions of the coded fields.

4.4. Frontier comparison and value added over the coded fields

Two frontier generative models were scored on the same 2,465 narrative-variable pairs against the same human labels with the same weights and metric code (Table 9). Calibration turns out to be a property of the individual model rather than of the paradigm, and that is the result which generalizes. The first frontier arm fails in the same direction as the typed model, rejected by the Spiegelhalter test at $z = -6.0$ and carrying a slope of 2.55 against 1.63, so both over-state prevalence. The second behaves differently, with a slope of 0.98, a statistic of -0.3 close to zero, and a calibration error of 0.0027 against 0.0282 and 0.0231 for the other two. It elicits only 79 distinct values against 87 for the first arm, so a coarse output grid is not by itself an obstacle to calibration. Calibration cannot therefore be inferred from the vendor or the interface, and the best-calibrated arm is not the most accurate one, which is why the audit cannot be skipped by buying a better model.

On accuracy the frontier arms do not agree with each other. The first reaches an F_1 of 0.967 against 0.908 for the typed model, a paired difference of 0.059 with an interval of [0.030, 0.098] over 2,416 judgments, which is a material advantage. The second reaches 0.885 and is not distinguishable from the typed model, because its paired difference of 0.023 has an interval of [-0.002, 0.048] that straddles zero, and it gets there by trading recall for precision, 0.814 against

0.970. The two frontier models differ from each other by 0.082, more than either differs from the typed model, so the choice among frontier models matters more than the choice between paradigms. Which variables a model fails on is idiosyncratic. At least one frontier arm recovers 3 of the typed model's 4 weak variables, the second arm falls below an F_1 of 0.80 on vehicle defect, aggression, wrong way, where the typed model is sound on 1, and the first is weak only on wrong way. This is the attribute-specific pattern that Bharati et al. (2026) report across six models, reproduced here against human labels rather than coded fields.

Cost is reported separately from accuracy, because the two cannot be measured on the same arms here. The benchmarked arms ran outside this pipeline and their outputs carry no token counts, so no statement in this paper attaches a cost to either of them. What can be priced is a scenario, applying the list prices of three commercially available tiers to the token counts this study measured. A mid-tier model would cost $7.95 per thousand narratives against $0.154 for the typed model, a factor of 52, with the largest tier at 129 and the smallest at 26. Projected to the whole corpus that is about $39,875 against $774, which separates a routine analysis from one that does not happen. The scenario prices tiers rather than the arms of Table 9, and the two must not be read as one curve.

Table 9
Jev against frontier generative models on the same human reference labels.

		Agreement with human labels				Calibration				
model	n	**prec.**	**rec.**	F_1	κ	**ECE**	**Brier**	**slope**	z	**prob.**
Jev 1.13	2,416	0.902	0.915	0.908	0.906	0.0231	0.0045	1.63	-5.3	native
claude-fable-5-1	2,416	0.959	**0.976**	**0.967**	**0.966**	0.0282	**0.0025**	2.55	-6.0	elicited
gpt-5.6-sol	2,416	**0.970**	0.814	0.885	0.883	**0.0027**	0.0056	0.98	-0.3	elicited

Note: All arms answer the same narratives and variables under the same criteria text, scored against the same labels with the same code. **Bold** is the leader per metric and bars scale F_1 from 0.80. The last column states whether the model returns a probability natively or was asked to state one. Every z is red because every arm is rejected, and they fail in opposite directions, since a slope above one with $z < 0$ over-states prevalence and a slope below one with $z > 0$ under-states it. Cost is not measured here, and Figure 7 prices a comparable tier at list rates. No arm is held out from the reference set.

Crash-level agreement with the coded fields runs from 97.9% to 99.7%, but at these base rates agreement is dominated by shared negatives, so Cohen's kappa is the informative summary (Table 10). It ranges from 0.10 for unbelted to 0.91 for animal involved, from slight to almost perfect on the Landis and Koch scale. Figure 6c plots how often each source flags a factor the other does not. Seven of the nine variables fall below the diagonal, so the narrative asserts these factors more often than the coded fields record them, and alcohol involvement sits on it. Restraint use is the one variable above, because the restraint field records an unbelted occupant in many crashes whose narrative says nothing about seat belts. That pattern is what the study was built to detect, and it is also why the coded fields cannot serve as an accuracy reference, since a narrative-only flag may be a model error or a coded omission.

Which of the two it is can only be settled on the human reference set. Its narratives that the model flags and the coded field records as negative are 41% of all its flags, 102 judgments over 9 variables. The coders confirm 59% of them, with an interval of 47 to 70%, against 99% where the coded field agrees. Rather more than half of the narrative-only region is therefore a factor the narrative states and the coded field does not record, which is a statement about what the two sources contain rather than about what happened in the crash. That rate divides sharply by the model's accuracy on the variable. Restricted to variables the model codes well it is 77% on 64 judgments, with an interval of 61 to 87%, and restricted to the weak variables it is 24% on 38, with an interval of 12 to 43%. Where the model is accurate a narrative-only flag is usually a factor the coded field omits, and where it is weak the flag is usually the model, so the value-added reading is licensed per variable by that variable's accuracy.

Against the same coded reference the typed model recovers more of the coded positives than keyword rules do, with precision moving in both directions (Figure 6b, Table 11). Recall rises for every variable, by the largest margins for hydroplaning and wrong-way travel, while precision rises for animal involvement, phone use and alcohol involvement and falls elsewhere, most for restraint use. The paired McNemar test of Eq. (9) rejects equal correctness for every variable except fatigue. Keyword rules therefore remain a strong and nearly free baseline where the vocabulary is closed, which matches the Arkansas finding of Bharati et al. (2026), and the typed model's advantage concentrates in variables whose expression is paraphrastic. Figure 7 places the two routes on the cost axis of the scenario above, with

keyword rules at a macro-averaged F_1 near one half and the typed model somewhat above it at a price two orders of magnitude higher.

Table 10
Discrepancy taxonomy against coded fields (agreement).

variable	n	agreement %	κ	both	narr. only	code only	neither
alcohol_involved	150,000	97.92	0.702	3,882	1,571	1,556	142,991
drug_involved	150,000	99.45	0.484	391	452	374	148,783
fatigue	150,000	99.26	0.707	1,369	729	387	147,515
animal_involved	150,000	99.67	0.910	2,565	459	37	146,939
phone_use	150,000	99.24	0.477	529	1,028	118	148,325
unbelted	150,000	97.99	0.103	187	707	2,307	146,799
hydroplane†	150,000	98.93	0.753	2,536	992	619	145,853
wrong_way	150,000	98.77	0.371	554	1,587	256	147,603
medical_episode	150,000	99.28	0.534	629	953	131	148,287

Note: Agreement against the incomplete coded fields rather than accuracy; see the note to Table 4. The dagger marks a coded field that is a proxy rather than a direct match. "Narrative only" is not an error rate, because a factor stated in the narrative and absent from the coded field is what this study sets out to find.

Table 11
Keyword baseline versus Jev on the same coded reference (agreement).

	Keyword rules			Jev 1.13			
variable	prec.	rec.	κ	prec.	rec.	κ	McNemar p
medical_episode	0.425	0.642	0.509	0.398	0.828	0.534	<0.001
drug_involved	0.576	0.284	0.378	0.464	0.511	0.484	<0.001
hydroplane	0.846	0.356	0.495	0.719	0.804	0.753	<0.001
fatigue	0.675	0.701	0.684	0.653	0.780	0.707	0.960
wrong_way	0.388	0.246	0.298	0.259	0.684	0.371	<0.001
animal_involved	0.751	0.860	0.798	0.848	0.986	0.910	<0.001
alcohol_involved	0.693	0.692	0.681	0.712	0.714	0.702	<0.001
phone_use	0.238	0.672	0.348	0.340	0.818	0.477	<0.001
unbelted	0.589	0.034	0.063	0.209	0.075	0.103	<0.001

Note: Agreement against the incomplete coded fields rather than accuracy; see the note to Table 4. McNemar tests paired per-narrative correctness. The frontier arm is scored against human labels in Table 9.

4.5. What changes in a safety diagnosis when narrative variables are added

The preceding subsections establish that the model reads narratives faithfully for most variables and that the coded contributing-factor fields omit a good deal of what the narratives state. Neither result says how much a transportation agency's picture of its own crashes would change if the narrative variables were used alongside the coded fields, and that question is the one a safety office actually asks. This subsection answers it on the Stage-2 random stratum, which contains 44,403 injury or fatal crashes of which 964 are fatal, using only data already produced and no further model calls.

The quantity is the number of injury and fatal crashes that would be attributed to each factor under each of two sources. Under the coded fields alone it is the count of such crashes whose coded field records the factor. Under the two sources together it is that count plus the expected number of crashes whose coded field is negative and whose narrative states the factor, obtained by summing the recalibrated probability of Section 4.2 over those crashes. Summing a calibrated probability rather than counting model flags is what makes the second term an estimate of a population total, because a calibrated probability already carries the model's error rate inside it. The recalibration map is fitted per

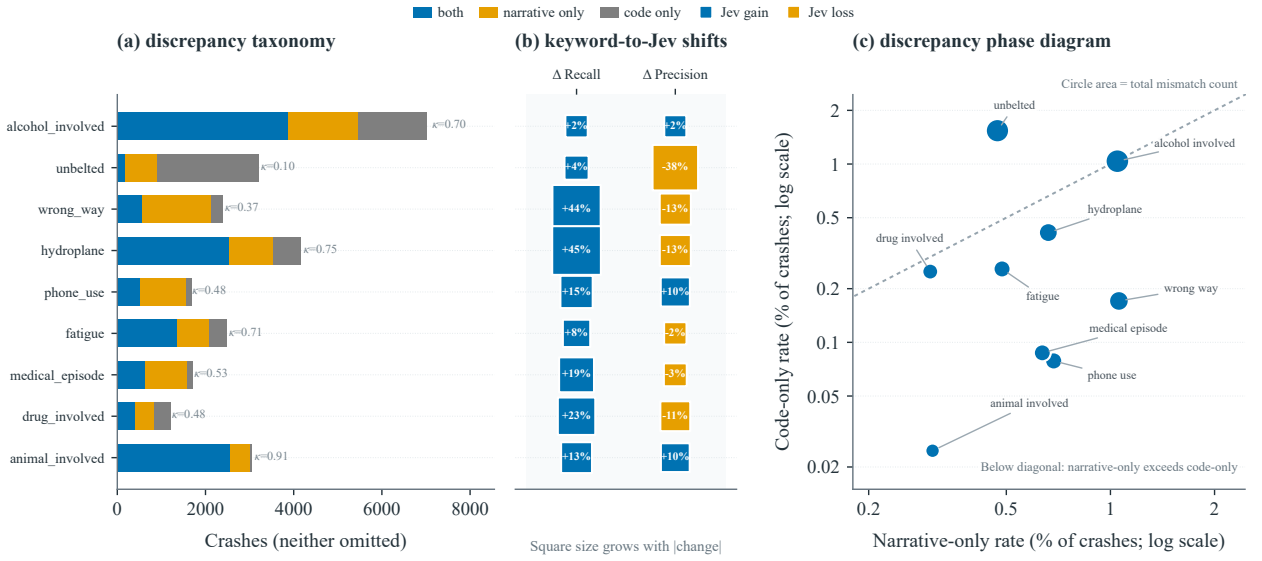


Figure 6: Agreement with the coded fields. (a) Discrepancy taxonomy per variable, with crashes flagged by both sources (blue), by the narrative only (orange) and by the code only (gray), and Cohen's kappa beside each bar. (b) The change in recall and in precision against the same coded reference when keyword rules are replaced by the typed model; the square grows with the size of the change, gains in blue and losses in orange. (c) The rate at which each source flags a factor the other does not, logarithmic on both axes, with the dashed diagonal marking equal rates and marker area the total mismatch count. Narrative-only cases are candidates for information the coded fields omit rather than errors.

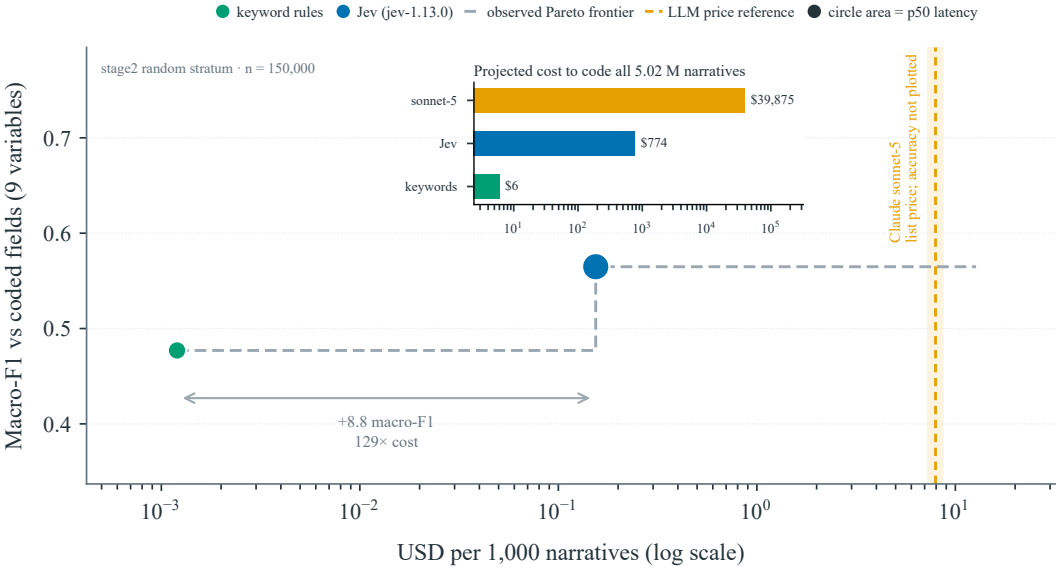


Figure 7: Cost per thousand narratives against macro-averaged F_1 with respect to the coded fields over the same nine variables, for keyword rules (green) and the typed model (blue), with marker area proportional to median latency and the dashed step the observed Pareto frontier. The vertical orange line is the list price of a mid-tier frontier model applied to the measured token counts. Frontier-arm accuracy is not plotted, because it was measured against the human reference set rather than the coded fields. The inset projects the cost of coding the full corpus by each route, which none of them coded.

variable wherever the reference set supports one. The interval on the second term comes from resampling the reference narratives and refitting the map, since the map is estimated on 400 narratives and the stratum holds 150,000, so the map is where the uncertainty lives.

Table 12 gives the result for all 9 mapped factors, scaled to Texas per year. The coded fields attribute 28,057 injury and fatal crashes per year across the nine factors, and the narrative variables add 10,747 more. The relative increase runs from 16% for unbelted to 182% for phone use, with a median of 39%. The largest single addition in absolute terms is 1,784 crashes per year for alcohol involved. The largest in relative terms is phone use, where the coded fields record 732 injury and fatal crashes per year and the narratives add 1,334 more, with an interval of 1,091 to 1,522. A factor that the coded fields make the smallest of the nine therefore becomes comparable to several of the others, which is a change in the diagnosis rather than only in the counts. Ranking the nine factors by attributed crashes, 4 of them change place when the narrative variables are added.

The confirmation rates of Section 4.4 decide how much of that change can be believed, and they differ enough across factors that the answer is not uniform. For 4 of the 9 factors a coder confirmed at least 0.70 of the narrative-only flags, and phone use is the strongest at 0.90, so for those the added count is evidence about the crashes rather than about the model. For 2 of them the coder confirmed fewer than half, and unbelted is the weakest at 0.15, so most of what the model adds there is model error and the increase is not a finding. The pattern is the one Section 4.3 would predict, since the factors with low confirmation are the factors the model codes poorly. A safety office can therefore adopt the narrative variables one factor at a time, on evidence it already holds, rather than as a single decision. Read that way, adding calibrated narrative variables to the coded fields raises the attributed burden of the well-measured factors by between 16% and 182%. phone use in particular moves from a minor entry in the coded record to a factor of a size that would ordinarily attract a countermeasure program.

Table 12
Injury and fatal crashes attributed to each factor, from the coded fields alone and with calibrated narrative variables added.

factor	coded field	coded per year	added by narratives per year	95% interval	increase %	narrative-only flags confirmed
alcohol_involved	coded_alcohol	8,531	1,784	[1,530, 2,649]	21	0.71
medical_episode	coded_medical	1,918	1,644	[1,371, 2,150]	86	0.60
phone_use	coded_phone	732	1,334	[1,091, 1,522]	182	0.90
wrong_way	coded_wrongway	1,543	1,248	[924, 2,791]	81	0.29
hydroplane†	coded_hydro_proxy	2,959	1,154	[932, 1,437]	39	0.88
unbelted	coded_unbelted	6,364	1,049	[800, 1,423]	16	0.15
drug_involved	coded_drug	1,833	998	[817, 1,317]	54	0.52
fatigue	coded_fatigue	2,476	910	[792, 1,020]	37	0.89
animal_involved	coded_animal	1,702	625	[509, 750]	37	–

Note: Counts are estimated on the Stage-2 random stratum and scaled to Texas per year. The added column is the expected number of injury or fatal crashes whose coded field is negative and whose narrative states the factor, summed over the recalibrated probability of Section 4.2, so it is a calibrated total rather than a count of flags. Intervals resample the reference narratives and refit the map. The last column is the share of narrative-only flags a coder confirmed, from Table 5, which disciplines the added column rather than feeding it. The dagger marks a coded field that is a proxy rather than a direct match.

5. Discussion

The central answer of this study is that adding calibrated narrative variables to a state's coded contributing-factor fields changes the safety diagnosis those fields support. Across the 9 factors that carry a coded counterpart, the narratives add 10,747 injury and fatal crashes per year to the 28,057 the coded fields record, a median increase of 39%, and 4 of the nine factors change rank. The strongest single case is phone use, which the coded fields make the smallest of the nine and which the narratives raise by 182%. That result rests on two prior ones that this paper also establishes. A typed decision model can code narratives at a cost low enough for a state to run the schema over a population rather than a sample. Its probabilities are usable for ranking but not for counting until they are recalibrated against a few thousand human judgments. Once that recalibration is in hand, the human review a narrative-derived

variable needs can be stated per variable and per year, which is what makes the added counts actionable rather than merely interesting. No comparable study has priced that change against the coded record. Safety diagnoses built on coded contributing factors alone, of the kind the crash data-quality literature describes (Imprialou and Quddus, 2019), therefore understate the factors whose evidence sits mainly in prose, and the size of the understatement is measurable rather than presumed. The closest prior study is the Arkansas benchmark of Bharati et al. (2026), which evaluated six frontier generative models on some four thousand matched fatal crashes over six attributes the database already codes. That study found a keyword baseline comparable to the best model, with differences across attributes larger than differences across models. The present results reproduce both findings on a different corpus and a different model class. Keyword rules remain a strong and nearly free baseline for closed-vocabulary variables, and the spread of kappa across variables against the coded fields is wider than any difference between methods. What this study adds is the layer that agreement measures cannot supply. It measures the probabilities rather than the labels, and it quantifies how much the administrative reference understates the model's fidelity to the narrative. It then converts the result into a review budget defined over the records an agency would read, which is the operational form of the human review that the Arkansas authors recommended. The paragraphs that follow take each of the study's findings in turn and place it against the closest published work.

The calibration finding sits between two literatures that have not been connected. Work on verbalized confidence in generative models reports over-confidence (Xiong et al., 2023) and shows that calibration depends on the elicitation format (Kadavath et al., 2022; Tian et al., 2023). Neither of the frontier arms audited here behaved in that way. The second arm elicited its probabilities on only 79 distinct values and was nonetheless the best calibrated of the three, at a slope of 0.98 and a statistic of -0.3, which says that a coarse verbalized scale is not by itself a source of over-confidence. The typed model and the first frontier arm failed in the other direction instead, with slopes above one and a level too high. That is the pattern the clinical prediction-model literature describes as a failure of calibration in the large that survives a reasonable slope (Van Calster et al., 2016; Steyerberg et al., 2010). The hierarchy of Van Calster et al. (2016) places mean calibration first, then weak calibration, which requires an intercept of zero and a slope of one together, then moderate and strong calibration. The typed model fails at the first level, with a mean probability of 4.8% against a prevalence of 2.7%, and it fails at the second as well, because its slope is 1.63 rather than one. Describing the remedy as a one-parameter correction would therefore be wrong. The Platt map of Eq. (20) fits an intercept and a slope, which is what a weak-calibration failure requires, and after it the pooled slope is 0.97. The result also extends the observation of Zhou et al. (2024) that larger and more instructable models become less reliable, because the more accurate frontier arm was the one rejected on calibration. Its slope of 2.55 lies further from one than the typed model's, even though its accuracy is higher. The arm that is best calibrated here, at a slope of 0.98 and a calibration error of 0.0027, is neither the most accurate arm nor the one with the finest output grid. Calibration is therefore a property of the model and the elicitation together rather than of size, price or interface, and an audit against an external reference is the only way to learn both its presence and its direction.

The reference-noise finding connects the injury-surveillance auto-coding line with the crash data-quality literature. The injury-surveillance auto-coding line classified narratives into cause codes with Bayesian and other learners (Lehto et al., 2009; Bertke et al., 2016) and routed low-confidence records to human review (Marucci-Wellman et al., 2015, 2017), and the rare-category study of Nanda et al. (2018) found that filtering rather than more training data rescued the rare classes. The present study inverts the direction of comparison, scoring the narrative-derived probability against the coded field and then against human judgment of the narrative. The result is not the one the design anticipated, and it is worth stating plainly. The two references give almost the same calibration error, a median ratio of 1.0 with a range of 0.8 to 2.2, so a coded reference does not flatter the model's probabilities. What an incomplete reference penalizes instead is the agreement statistic. Cohen's kappa against the coded field falls below the weighted kappa against human judgment for all 9 comparable variables, by a median of 0.26, and drug involved reads as 0.48 against the coded record and 0.96 against human judgment of the same narratives. The mechanism is omission in the coded fields, whose incompleteness the crash data-quality literature lists among its most serious problems (Imprialou and Quddus, 2019), and which this study measures directly as the excess of narrative-only over code-only discrepancies. Under-reporting studies have measured the same omission at the level of whole crashes and injured people (Elvik and Mysen, 1999; Alsop and Langley, 2001; Watson et al., 2015). Lemma 2 gives that structure its formal statement, and its empirical content here is narrower than the lemma allows. Omission moves the agreement statistic rather than the calibration statistic, because a probability close to the rate at which a factor is stated stays close to it whether or not a clerk recorded the factor. Arteaga and Park (2025) used a language model to expose under-reporting from the same kind of narrative evidence. The confirmed rate of 59% in the narrative-only cell reported here measures how much of

that apparent under-reporting is a factor the coded field omits rather than a model error. Its split by variable accuracy shows that the answer depends on the model as much as on the field, and Section 4.5 converts it into crashes per year.

The resolution floor is a general property of any model that reports probabilities on a discrete grid, and the calibration literature has not treated it because that literature is written for continuous outputs. Reliability diagrams and their binning have received close attention, from the consistency bands of Bröcker and Smith (2007) to the binning-free constructions of Dimitriadis et al. (2021) and the regression view of Gneiting and Resin (2023). Kumar et al. (2019) showed that verifiable calibration error requires a discretized output, and a model that discretizes its own output supplies that property for free. The exact error of Eq. (12) therefore needs no binning parameter, and equal-mass binning of the kind recommended by Nixon et al. (2019) stops working because the mass concentrates on a handful of attainable values. Proposition 1 adds the constraint that a discrete grid imposes. Wherever the smallest positive value exceeds a variable's base rate, no assignment of grid values is calibrated and the mean probability over-states prevalence by at least the gap. That check takes one line, and any analyst using such a model should run it before trusting summed probabilities. The floor was invisible in the developer's documentation, and it was found only because the audit inspected the distribution of returned values rather than their accuracy alone.

The review budget over flagged records connects selective prediction to rare-event screening. The risk-coverage machinery of El-Yaniv and Wiener (2010) and Geifman and Yaniv (2017) assumes that abstention trades coverage against error, and on a balanced problem it does. On a variable with a base rate of a few percent, a model that answers everything already has an error rate below conventional targets, purely because the negative class dominates. The analysis then certifies a system nobody should trust, which is the objection the precision-recall literature raised against error rates on imbalanced data (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015). Pointing the construction at the decision actually being made, which is whether to read a flagged record, restores the trade-off. Nobody reviews the records a model calls negative, so defining coverage over flagged positives and risk as one minus precision produces a number an agency can staff against. The difference between the coded-field budget, which saturates for eight of nine variables, and the human-referenced budget of 0.8% is the reference effect seen in the agreement statistics, now expressed in hours. A budget selected and certified on the same labels is optimistic, which is why the value reported here is cross-fitted and carries a lower bound on the precision it delivers. That bound is a finite-sample statement about this reference set rather than a guarantee. The distribution-free risk-control literature supplies guarantees of the stronger kind, with risk-controlling prediction sets (Bates et al., 2021), conformal risk control (Angelopoulos et al., 2024) and the Learn then Test framework for selecting among many candidate thresholds (Angelopoulos et al., 2025). Applying that machinery here would require an exchangeable calibration sample and a fixed error level chosen in advance, and it would return a threshold whose risk is controlled with high probability rather than an estimate with an interval. Section 4.3 reports the latter, and the construction is not specific to crash narratives, since it applies to any screening task in which the positive class is rare and the cost is concentrated in confirming positives.

The schema and the cost model transfer to other states more readily than the reference mapping does. The schema asks about the narrative rather than about Texas, so the same questions can be posed to any state's narrative field, with the criteria text serving as the complete specification of each variable for a model, a coder or a reviewer alike. The cost model transfers more strongly still, because it is dominated by schema size rather than by narrative length. The per-narrative cost of this schema is therefore approximately the same in any corpus of comparable prose length, and the break-even fraction of Eq. (4) is a property of the schema. What does not transfer is the coded-field reference. Every state codes contributing factors differently, and the mapping of Table B.2 had to be corrected twice against the actual value sets before it was usable. The one-sided error that Lemma 2 describes will also have a different rate in every field of every state. An agency adopting the pipeline should therefore expect to reuse the schema and to refit the recalibration map on its own human judgments. It should also expect to rebuild the reference mapping from its own data dictionary, and the released code is organized so that those three steps are separable.

Transfer is a hypothesis here rather than a finding, because this study covers one state, one model version and one schema, and it is stated with the conditions under which it would fail. The cost model would fail where narratives are much longer than the 378 characters of the median here, since the text term would then stop being small beside the schema term. The schema would fail where officers write to a different convention, because the criteria text was written against Texas shorthand and a construct such as an intentional act is recognized through the phrases a reporting culture happens to use. The recalibration map would fail wherever the prevalence differs, since a map fitted at one base rate moves the probabilities toward that rate and not toward another. The accuracy results would fail on any later model version, because the audit measures a model and a schema together rather than a capability. Each of these is testable

in one state with a few hundred labeled narratives, which is the cost this study reports, and until that test is run the transfer claim should be read as a design expectation.

6. Conclusion

This study set out to answer two questions that crash-narrative research has left to the reader, namely how much human checking a narrative-derived variable needs and what changes in a safety diagnosis when such variables are added to the coded record. Answering them required treating narrative coding as a set of typed, gated decisions with probabilities attached and auditing the probabilities against an external reference instead of assuming them. It also required running that decomposition at scale, with a first-stage screen over 499,500 narratives and full coding of 195,857 rather than a few thousand records. The typed model coded those 195,857 Texas narratives with a 27-question schema at \$0.154 per thousand, which projects to \$774 for the full corpus of 5,018,080 narratives at the same rate. The cost model showed that spend is governed by the size of the question schema rather than by the length of the text, which makes batching questions the dominant efficiency lever and sets a break-even fraction of 0.67 below which two-stage screening pays. The model returns probabilities on a two-decimal grid whose floor exceeds the coded-field base rate of 4 target variables, so for those variables no allocation of grid values can be calibrated and the mean probability is not a prevalence estimate. And risk-coverage analysis imported unchanged from balanced settings certifies rare-event screeners as needing no review, so the budget has to be defined over the records the model flags.

Measured against human judgment of the same criteria, the model reaches an F_1 of 0.908 and a weighted kappa of 0.906 over 2,416 usable consensus labels, but its probabilities are not calibrated. The Spiegelhalter statistic is -5.3 and the mean probability lies above the prevalence, so the model fails calibration in the large, and its slope of 1.63 means it fails weak calibration as well. The two references agree on the calibration error, at a median ratio of 1.0, and they disagree on agreement, where the coded field reads lower than human judgment for every comparable variable by a median of 0.26 in kappa. The coded reference is incomplete in one direction, and what that incompleteness costs is the appearance of agreement rather than the appearance of calibration. The same 2,416 judgments then recalibrate the model to an out-of-fold calibration error of 0.0069 with a slope of 0.97, which is the operationally important result. For a few thousand judgments a safety office obtains probabilities it can sum, threshold and budget against over a corpus it could never label by hand. Adding those probabilities to the coded contributing-factor fields raises the injury and fatal crashes attributed to the nine mapped factors by 10,747 per year and moves 4 of them in rank, which is the answer to the second question. The frontier comparison showed that two generative models from different vendors do not agree with each other about how much better than the typed model they are. It also showed that calibration is a property of the individual model rather than of the paradigm, so the transferable part of this work is the audit rather than the model. The contributions are therefore a gated typed decomposition whose design rules were measured, a large-scale run with a measured cost model, and an independent calibration and selective-prediction audit against two references. To these the paper adds a review budget defined over flagged records and certified out of fold, a downstream measurement of what calibrated narrative variables change in a safety diagnosis, and the released schema, code and outputs.

The study has boundaries that the reader should hold in mind, and this paragraph states them. The model is closed and commercial, its training data are undisclosed, and its behavior may change without notice. The pinned identifier and the recorded per-call identifier make such a change detectable but not preventable, so a reader cannot rerun the analysis if the vendor withdraws the model, and audits of commercial systems should be treated as perishable. The coded CRIS fields used as the large reference are incomplete, so every statistic computed against them is an agreement measure bounded by the reference's own error. The accuracy claims rest on a human reference set that is smaller by three orders of magnitude and that is defined only on cases the coders found clear, because disputes were excluded rather than adjudicated. Its pairwise reliability is estimated on 94 triple-coded pairs, and its stratification on the model's probability rather than on the positive class leaves the rarest variables with a handful of positive labels, as few as 5 for 2 of the 4 that fail. Per-variable precision, any threshold fitted to it, and the calibration error at the floor stratum are correspondingly imprecise for those variables, and 5 variables carry too few positives to support a recalibration map of their own. The weights that make the reference set speak for the population are calibration weights rather than known inclusion probabilities, because the frame was drawn by a greedy cell-filling rule and then screened. Their validity therefore rests on labeling being uninformative about the outcome within a cell. The recalibration maps were fitted on 3 coders' judgment of Texas narratives and should be refitted rather than transported. The downstream counts of Section 4.5 inherit every one of these boundaries, and for the 2 factors whose narrative-only flags the coders confirmed less than half the time they should be read as an upper bound rather than as an estimate. The frontier arms ran with more

context and more computation per judgment than the typed model, and their cost was not metered at all, so no cost in this paper describes them and the price comparison is a scenario over commercially listed tiers. The analysis covers one state, narratives are written by officers with agency-specific conventions and are themselves incomplete records of the crash, and residual identifiers survive the data owner's de-identification pass at the rates reported in Section 3.1, which is why no narrative is reproduced here.

Four lines of work follow from those boundaries. The first is a case-by-case adjudication of the discrepancies between narrative-derived and coded values, sampled on the discrepancies themselves rather than on a probability-stratified frame. That design would narrow the wide intervals on the narrative-only cell and attribute each discrepancy to the narrative, the code or the model. The second is a revised schema for the 4 variables that failed, namely unbelted, intentional, wrong way, vehicle defect, with their criteria rewritten after error analysis. The revision would be scored against a fresh reference set drawn against the revised schema, so that the repair is validated on labels that did not select it. The third is replication in other states, which would test whether the schema and the cost model transfer as Section 5 predicts. The fourth is a repeated audit across model versions, with the same reference set and the same metric code. It is the only way to learn whether the calibration and the review budgets reported here are properties of the model or of one release, and the released code is designed to make it routine.

Appendices

A. The question schema

Table A.1 reproduces every question in the schema as it was sent to the model, with its type, its gate where it has one, and the complete instruction and criteria text. The text is the entire specification of each variable, because the model sees nothing else, and the human coders of Section 3.3 and the frontier models of Section 4.4 received the same text verbatim. The schema was produced in two steps that the released files record. A first version was written from the design rules of Section 3.2 and run on a development sample of 2,000 narratives, and the error analysis of that run produced the exclusion clause of the aggression question, whose effect is reported in Section 3.2. The file reproduced here is the second version, which is the one used for the second-stage run, the reference set and the frontier arms, and it is released with the code together with the lean first-stage schema of 10 presence questions. Instructions refer to the narrative by its key in the state object, and criteria for a presence question name the boundary cases that count and do not count, which is where most of the design effort went. The residual-identifier question is included in the file because it ships with the schema, but it was run as a separate display screen and was not part of the population coding.

Table A.1
The full question schema, version 1.1, as sent to the model.

id	type	gate	instruction and criteria
`preg_mentioned`	noul	–	The 'narrative' states that a person involved in the crash was pregnant. *true*: Any explicit statement that a driver, passenger, pedestrian, or other involved person is pregnant, e.g. 'was 6 months pregnant', 'pregnant driver', 'expecting'. *false*: No pregnancy is stated. Mentions of children, infants, or car seats do not count.
`medical_episode`	noul	–	The 'narrative' states that a driver experienced a medical event (seizure, fainting, blackout, diabetic episode, heart attack, stroke, or similar) before or during the crash. *true*: A medical event of the driver is described as occurring before or during the crash, whether confirmed or suspected by the officer. *false*: No medical event of the driver is described. Injuries caused by the crash, or transport to a hospital for crash injuries, do not count.

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Table A.1, continued

id	type	gate	instruction and criteria
drug_involved	noul	–	The 'narrative' states or suspects that a driver was under the influence of drugs or medication (not alcohol). *true*: Any statement that a driver used, was impaired by, or was suspected of using illegal drugs, marijuana, prescription medication, or other substances other than alcohol. *false*: No drug or medication involvement of a driver is mentioned. Alcohol-only impairment does not count.
alcohol_involved	noul	–	The 'narrative' states or suspects that a driver had been drinking alcohol or was intoxicated by alcohol. *true*: Any statement that a driver had been drinking, smelled of alcohol, was DWI or DUI, or was suspected of alcohol intoxication. *false*: No alcohol involvement of a driver is mentioned.
hydroplane	noul	–	The 'narrative' states that a vehicle hydroplaned or lost traction because of water on the roadway. *true*: The word hydroplane appears, or the narrative says a vehicle lost traction, skidded, or lost control because of rain, wet roadway, or standing water. *false*: No loss of traction due to water is described. Wet conditions mentioned without loss of control do not count.
vehicle_defect	noul	–	The 'narrative' states that a mechanical failure of a vehicle (tire blowout, brake failure, steering failure, or similar) contributed to the crash. *true*: A tire blowout, tread separation, brake failure, steering failure, stuck accelerator, or other mechanical defect is described as contributing to the crash. *false*: No mechanical failure is described. Damage caused by the crash does not count.
fatigue	noul	–	The 'narrative' states that a driver fell asleep, dozed off, was drowsy, or was fatigued. *true*: A driver is described as asleep, falling asleep, nodding off, drowsy, tired, or fatigued before the crash. *false*: No sleep or fatigue of a driver is mentioned.
aggression	noul	–	The 'narrative' describes road rage, aggressive driving directed at another road user, or a physical or verbal confrontation between people involved. *true*: Road rage, brake-checking, chasing, threatening, arguing, fighting, or a confrontation before or after the crash is described. *false*: No aggression or confrontation is described. Ordinary traffic violations without hostility do not count. A police pursuit, a traffic stop, an officer chasing or stopping a vehicle, and a driver evading or fleeing from police do not count. A driver leaving the scene of the crash does not count by itself.
intentional	noul	–	The 'narrative' states or suspects that a vehicle was used intentionally to cause the crash (assault with a vehicle, deliberate ramming, or a suicide attempt). *true*: The crash is described as deliberate, intentional, on purpose, an assault with a vehicle, or a suicide attempt. *false*: The crash is described as unintentional, or intent is not mentioned.
hit_and_run	noul	–	The 'narrative' states that a driver left the scene of the crash without stopping to provide information. *true*: A driver fled, left the scene, failed to stop and render aid, or drove away without exchanging information. *false*: All drivers remained at the scene, or leaving the scene is not mentioned.

continued on the next page

Table A.1, continued

id	type	gate	instruction and criteria
`animal_involved`	noul	–	The 'narrative' states that an animal on or near the roadway was involved in the crash or in the driver's evasive action. *true*: A vehicle struck an animal, or swerved or braked to avoid an animal. *false*: No animal is mentioned as involved.
`wrong_way`	noul	–	The 'narrative' states that a vehicle was traveling the wrong way on a roadway, ramp, or lane designated for the opposite direction. *true*: A vehicle is described as going the wrong way, wrong direction, or against traffic on a one-way road, divided highway, or ramp. *false*: No wrong-way travel is described. Crossing the center line during a loss of control does not count.
`phone_use`	noul	–	The 'narrative' states that a driver was using a phone or mobile device at the time of the crash. *true*: A driver is described as texting, talking on a phone, looking at a phone, or using a mobile device. *false*: No phone or device use is mentioned. General 'distracted' or 'inattention' without a device does not count.
`unbelted`	noul	–	The 'narrative' states that a vehicle occupant was not wearing a seat belt. *true*: An occupant is described as unrestrained, not wearing a seat belt, or unbelted. *false*: Seat belt use is not mentioned, or occupants are described as belted.
`transported`	noul	–	The 'narrative' states that a person was transported to a hospital or medical facility. *true*: A person was transported by EMS, ambulance, helicopter, or private vehicle to a hospital, medical center, or clinic. *false*: No transport to a medical facility is mentioned, or people refused transport.
`witness_cited`	noul	–	The 'narrative' cites a statement from a witness who was not a driver or occupant of the involved vehicles. *true*: A witness, bystander, or independent observer's statement is described. *false*: Only driver, passenger, or officer statements appear, or no statements appear.
`medical_type`	choice	`medical_episode`	Which type of driver medical event does the 'narrative' describe as occurring before or during the crash? *seizure*: Seizure or epileptic episode. *loss_of_consciousness*: Fainting, passing out, blacking out, syncope, or becoming unresponsive without a stated cause. *diabetic*: Diabetic episode, low blood sugar, hypoglycemia, or insulin-related event. *cardiac*: Heart attack, cardiac arrest, chest pain, or other heart event. *stroke*: Stroke or stroke-like symptoms. *other_medical*: Another medical condition or episode of the driver, such as dementia, dizziness, or an unspecified medical emergency. *none*: No driver medical event is described.
`drug_class`	choice	`drug_involved`	Which category of drug or medication does the 'narrative' associate with a driver? Choose based only on what the narrative names. *marijuana*: Marijuana, THC, cannabis, or weed. *stimulant*: Methamphetamine, cocaine, amphetamines, or other stimulants. *opioid*: Heroin, fentanyl, oxycodone, hydrocodone, or other opioids. *sedative_prescription*: Xanax, benzodiazepines, sleeping pills, muscle relaxers, or other prescription sedatives. *unspecified_drug*: Drugs, narcotics, a controlled substance, or medication mentioned without naming the type. *none*: No drug or medication is mentioned.

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Table A.1, continued

id	type	gate	instruction and criteria
`preg_role`	choice	`preg_mentioned`	What was the role of the pregnant person in the crash according to the 'narrative'? *driver*: The pregnant person was driving one of the vehicles. *passenger*: The pregnant person was a passenger in one of the vehicles. *pedestrian_or_other*: The pregnant person was a pedestrian, cyclist, or otherwise not inside a vehicle. *no_pregnancy*: The narrative does not state that anyone was pregnant.
`preg_outcome`	choice	`preg_mentioned`	What does the 'narrative' state about the condition of the pregnant person after the crash? *no_complaint*: The pregnant person reported no injury or the narrative gives no complaint. *pain_or_evaluation*: The pregnant person complained of pain, tightness, or discomfort, or was checked by EMS but not transported. *transported*: The pregnant person was transported to a hospital or medical facility as a precaution or for injury. *fetal_harm*: The narrative states harm to, or loss of, the unborn child. *no_pregnancy*: The narrative does not state that anyone was pregnant.
`preg_stage`	choice	`preg_mentioned`	What stage of pregnancy does the 'narrative' state for the pregnant person? *early*: Up to 13 weeks, or 1 to 3 months pregnant, or described as first trimester or early pregnancy. *mid*: 14 to 27 weeks, or 4 to 6 months pregnant, or described as second trimester. *late*: 28 weeks or more, or 7 to 9 months pregnant, or described as third trimester, due soon, or full term. *stage_not_stated*: Pregnancy is stated but no weeks, months, or trimester is given. *no_pregnancy*: The narrative does not state that anyone was pregnant.
`animal_type`	choice	`animal_involved`	Which kind of animal does the 'narrative' describe as involved in the crash? *deer*: Deer. *dog*: Dog. *livestock*: Cow, cattle, horse, goat, or other livestock. *feral_hog*: Hog, feral hog, wild boar, or pig. *other_animal*: Any other animal, such as coyote, cat, or an unnamed animal. *none*: No animal is described as involved.
`defect_type`	choice	`vehicle_defect`	Which kind of vehicle mechanical failure does the 'narrative' describe as contributing to the crash? *tire*: Tire blowout, flat tire, or tread separation. *brakes*: Brake failure or brakes not working. *steering_or_suspension*: Steering failure, wheel came off, axle or suspension failure. *other_mechanical*: Another mechanical failure such as engine stall, stuck accelerator, or trailer hitch failure. *none*: No mechanical failure is described.

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Table A.1, continued

id	type	gate	instruction and criteria
`surface_narr`	choice	–	What roadway surface condition does the 'narrative' state at the time of the crash? *dry*: The roadway is described as dry. *wet*: The roadway is described as wet, rainy, or having standing water. *ice_or_snow*: The roadway is described as icy, frozen, slick from ice, or snowy. *other_surface*: Another surface hazard is described, such as gravel, oil, mud, sand, or debris. *not_stated*: The narrative does not describe the surface condition.
`narrative_detail`	score	–	How much crash-relevant detail does the 'narrative' contain? *level 0*: Boilerplate or a single sentence with no description of how the crash happened. *level 1*: A basic sequence of vehicle movements and the collision, without statements or contributing factors. *level 2*: A detailed account including driver or witness statements, contributing factors, or officer findings.
`emotion_intensity`	score	–	How intense is the emotional state of any driver described in the 'narrative' before or after the crash? *level 0*: No emotional state is described. *level 1*: A driver is described as upset, frustrated, crying, shaken, or scared. *level 2*: A driver is described as angry, hostile, yelling, threatening, or violent.
`injury_narr`	score	–	What is the most severe injury outcome stated in the 'narrative' for any person? *level 0*: No injury is mentioned, or the narrative says no one was injured. *level 1*: A person complained of pain or possible injury but was not transported. *level 2*: A person had a visible injury or was transported to a hospital. *level 3*: A person was pronounced dead or the narrative describes a fatality.
`pii_residual`	noul	–	The 'narrative' contains a person's full name, date of birth, driver license or ID number, home address, or phone number that is not already replaced by a placeholder. *true*: An actual name, date of birth, license or ID number, street address, or phone number appears as literal text. *false*: No such identifier appears, or every identifier has been replaced by a placeholder such as [NAME] or [PHONE].

Note: Text is reproduced from `schemas/crash_factors_v1_1.json` without abbreviation. A gated question is interpreted only when its gate exceeds $\tau_g = 0.5$, otherwise it takes its no-match option. `pii_residual` ships in the same file but was run separately as the display screen, outside the Stage 1 and Stage 2 calls.

B. Keyword families and the coded-field mapping

The rule baseline of Section 3.3 is a set of case-insensitive regular expressions, one family per variable, and Table B.1 lists them as they appear in the scripts that apply them. The first ten families were written to over-find rather than to be precise, because their original purpose was to enrich the rare classes of the second-stage frame and to provide a population keyword rate over all 5,018,080 narratives. They were tuned only on the development sample and never on any evaluated narrative. The last three families, for alcohol involvement, phone use and restraint use, were added so that the baseline covers every variable that has a coded reference, since comparing a macro-averaged score over different variable sets would not be a comparison. Those three were scored on the working set of narratives only. The residual-identifier patterns of Section 3.1 are of the same kind and are listed in the released code beside them. Table B.2 gives the mapping from each narrative variable to the coded CRIS fields that define its reference, which is implemented in the released join script over the crash-unit-person extract. Each reference is true for a crash when the rule holds on any person or unit row of that crash, because a contributing factor is recorded per unit and a restraint or

test result per person. The hydroplaning reference is a proxy built from the loss-of-control, surface and weather codes, because CRIS has no hydroplaning field. The final row is the opposite-direction collision type that the original plan proposed for wrong-way travel, retained only as a sensitivity check after it proved a factor of 17 too broad.

Table B.1
Regular-expression families of the rule baseline.

family	narrative variable	scope	regular expression (case-insensitive)
preg	preg_mentioned	all narratives	`pregnan|unborn|fetus|fetal|trimester|weeks along|miscarr`
medical	medical_episode	all narratives	`seizure|diabet|passed out|blacked out|fainted|heart attack|stroke|unconscious|medical (episode|condition|emergency|event)|dementia|alzheimer`
drug	drug_involved	all narratives	`marijuana|methamphetamine|meth |cocaine|fentanyl|xanax|heroin|pills|narcotic|controlled substance|under the influence of (a )?drug`
hydro	hydroplane	all narratives	`hydroplan`
defect	vehicle_defect	all narratives	`tire blow|blowout|blew out|brake fail|brakes fail|mechanical fail|steering fail`
rage	aggression	all narratives	`road rage|angry|anger|enraged|argument|fight|confront|brandish|assault`
fatigue	fatigue	all narratives	`fell asleep|asleep|drowsy|fatigue|nodded off`
intent	intentional	all narratives	`suicid|intentional|deliberate|on purpose|rammed`
wrongway	wrong_way	all narratives	`wrong way|wrong direction`
animal	animal_involved	all narratives	`deer|hog|cow|horse|dog|coyote|cattle`
alcohol	alcohol_involved	working set	`\balcohol|intoxicat|\bdwi\b|\bdui\b|drunk|\bbac\b|been drinking|odor of an alcoholic|open container|breathalyz|field sobriety`
phone	phone_use	working set	`cell ?phone|cellular|mobile (phone|device)|texting|\btext(ed|ing)?\b|on (the|his|her|their) phone|looking at (his|her|their) phone`
unbelted	unbelted	working set	`not wearing (a )?(seat ?belt|safety belt)|unbelted|unrestrained|no (seat ?belt|restraint)|seat ?belt was not|without a seat ?belt`

Note: The first ten families apply to all narratives and provide the population keyword rates of Table 3. The last three were added so the baseline covers every variable with a coded reference, and were scored on the working set only. Patterns are read from the scripts that apply them.

Table B.2
Mapping from narrative variables to coded CRIS reference fields.

narrative variable	reference column	CRIS fields and values	aggregation to crash level
alcohol_involved	`coded_alcohol`	`Contrib_Factr_1_ID` or `Contrib_Factr_2_ID` matches "Under Influence - Alcohol" or "Had Been Drinking"; or `Prsn_Alc_Rslt_ID` = 1; or `Prsn_Bac_Test_Rslt` > 0	any person or unit row of the crash
drug_involved	`coded_drug`	contributing factor matches "Under Influence - Drug"; or `Prsn_Drg_Rslt_ID` = 1	any row

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Table B.2, continued

narrative variable	reference column	CRIS fields and values	aggregation to crash level
fatigue	`coded_fatigue`	contributing factor matches “Fatigued Or Asleep”	any row
medical_episode	`coded_medical`	contributing factor matches “Ill (Explain In Narrative)”	any row
animal_involved	`coded_animal`	contributing factor matches “Animal On Road”; or `Harm_Evnt_ID` contains “Animal”	any row
phone_use	`coded_phone`	contributing factor matches “Cell/Mobile Device Use”	any row
unbelted	`coded_unbelted`	`Prsn_Rest_ID` equals “None” (the only unbelted code; “Not Applicable” excluded)	any person row
hydroplane	`coded_hydro_proxy`	`Othr_Factr_ID` contains “Lost Control Or Skidded” and (`Surf_Cond_ID` contains “Wet” or “Standing Water”, or `Wthr_Cond_ID` contains “Rain”)	any row; proxy reference
wrong_way	`coded_wrongway`	contributing factor matches “Wrong Way - One Way Road”, “Wrong Side - Not Passing” or “Wrong Side - Approach”	any row
(sensitivity only)	`coded_wrongway_proxy`	`FHE_Collsn_ID` contains “Opposite Direction”	any row; not used as a reference

Note: Each reference is true for a crash when the rule holds on any of its person or unit rows. The hydroplaning reference is a proxy built from surface, weather and loss-of-control codes, because CRIS has no hydroplaning field. The final row is the opposite-direction collision type the original plan proposed for wrong-way travel, retained only as a sensitivity check.

C. Coder protocol and additional reliability diagrams

The human reference set of Section 3.3 was produced under a written protocol that this appendix summarizes, and Figures C.1 and C.2 draw the two human-referenced diagrams that the main text refers to. The protocol defined the task as one narrow question repeated, whether the narrative states the factor, under the governing rule to label what the narrative states rather than what probably happened. A crash that an experienced analyst would read as alcohol-involved is therefore labeled negative if the officer did not write it down. That rule inverts the usual assumption that more domain expertise is better, because the dominant risk is over-inference. Coders who know crashes may fill in what the officer omitted, agree with each other, and produce a reference that is confidently wrong in the direction the model might also be wrong. The panel was therefore mixed by design, with one crash-data practitioner who reads officer shorthand, one transportation-safety researcher familiar with the constructs, and one careful non-specialist reader who cannot over-infer and who serves as the check on the other two. Coders were required to be independent of the modeling work, never to have seen the model’s outputs or helped write the criteria, and not to discuss items while coding. All three first coded a shared calibration block, which was reviewed together once to catch inference, ambiguous criteria and shorthand the non-specialist could not read, after which no further discussion took place. Each coder then labeled an allocation of narratives determined by position in a fixed roster, with every narrative coded by two of the three and the calibration block by all three. The roster rule was deterministic and fixed before any narrative was seen. Narratives were held in the order the export produced, and the calibration block occupied the first positions and was assigned to every coder. Each later position went to the two coders whose index matched that position under a fixed rotation, so no coder chose a narrative and no narrative was reassigned after labeling began. Each judgment carried the coder identifier, the time taken and a yes, no or unclear verdict. Reliability was computed from the double- and triple-coded pairs as reported in Section 3.3, majority labels were formed, and ties and unclear verdicts were recorded as disputed and excluded.

The calibration weights of Section 3.3 are built from the frame rather than from the labels, and their construction is recorded here in full. The population is the second-stage random stratum. For each of the 16 presence variables the population is divided into the cells [0,0.05); [0.05,0.3); [0.3,0.7); [0.7,0.95); [0.95,1] and the labeled pairs falling in each cell are scaled so that their weights sum to the population count of that cell. 1,265 of the pairs entered through the active route, with assignment probability one, and 1,200 through the control route, with assignment probability 3 divided by the narrative’s count of low-probability variables. Cells holding fewer than 5 labeled pairs were merged with an adjacent cell of the same variable before the scaling, which affected 6 variables, and the merges are listed in the released design report. The weights span a factor of 189 and give a design effect of 1.95, with per-variable effective

sample sizes running from 64 to 93. The cluster bootstrap resamples narratives within the 5 draw routes the frame records, which are the cells the greedy rule drew each narrative to fill.

Table C.1
Unresolved pairs and the sensitivity of the headline results to them.

block	item	count	F_1	ECE	$H(0.90)$ %
by variable	hydroplane	9	–	–	–
by variable	hit_and_run	7	–	–	–
by variable	phone_use	7	–	–	–
by variable	transported	6	–	–	–
by variable	witness_cited	4	–	–	–
by variable	alcohol_involved	4	–	–	–
by variable	vehicle_defect	3	–	–	–
by variable	aggression	3	–	–	–
by variable	medical_episode	2	–	–	–
by variable	wrong_way	2	–	–	–
by variable	fatigue	1	–	–	–
by variable	drug_involved	1	–	–	–
by probability bin	[0.05,0.3)	9	–	–	–
by probability bin	[0.3,0.7)	22	–	–	–
by probability bin	[0.7,0.95)	17	–	–	–
by probability bin	[0.95,1]	1	–	–	–
sensitivity	excluded, as reported throughout	49	0.908	0.0231	0.0
sensitivity	counted as model errors	49	0.875	0.0228	6.9
sensitivity	counted as model-correct	49	0.912	0.0232	0.0
sensitivity	routed to review regardless of probability	49	–	–	3.7

Note: The 49 excluded pairs are those the coders marked unclear or disagreed on. They are dropped from every rate in Section 4.3, so those rates are conditional on the pairs the coders found clear. The sensitivity block replaces that conditioning with three assumptions. Routing the excluded pairs to review leaves F_1 undefined, because a record sent to a human is not scored as a model decision.

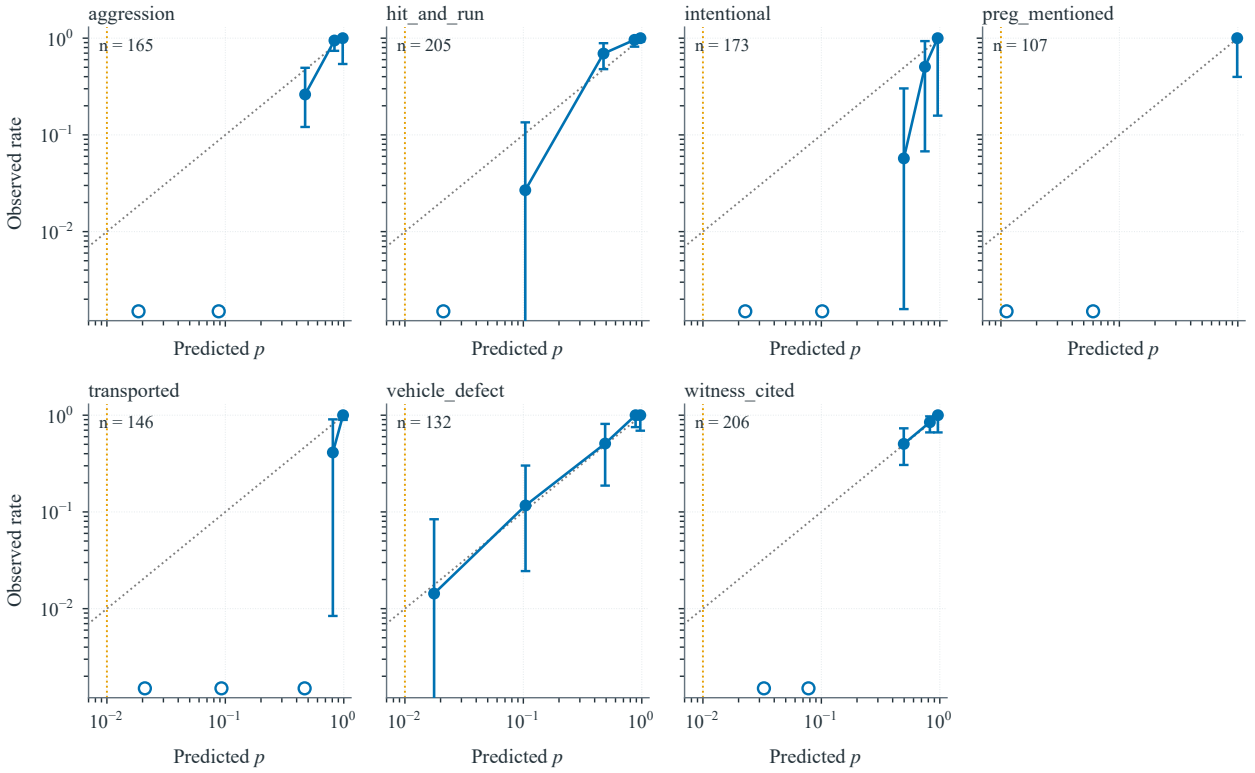


Figure C.1: Reliability against the human reference set for the seven presence variables with no coded counterpart, which therefore do not appear in Figure 4. Each panel gives the observed positive rate among the human labels within each sampling stratum, weighted by the calibration weights of Section 3.3, with approximate intervals on the Kish effective sample size, on logarithmic axes. The dotted diagonal marks calibration, the vertical guide the grid floor, open circles strata with no positive, and the number of judgments is printed.

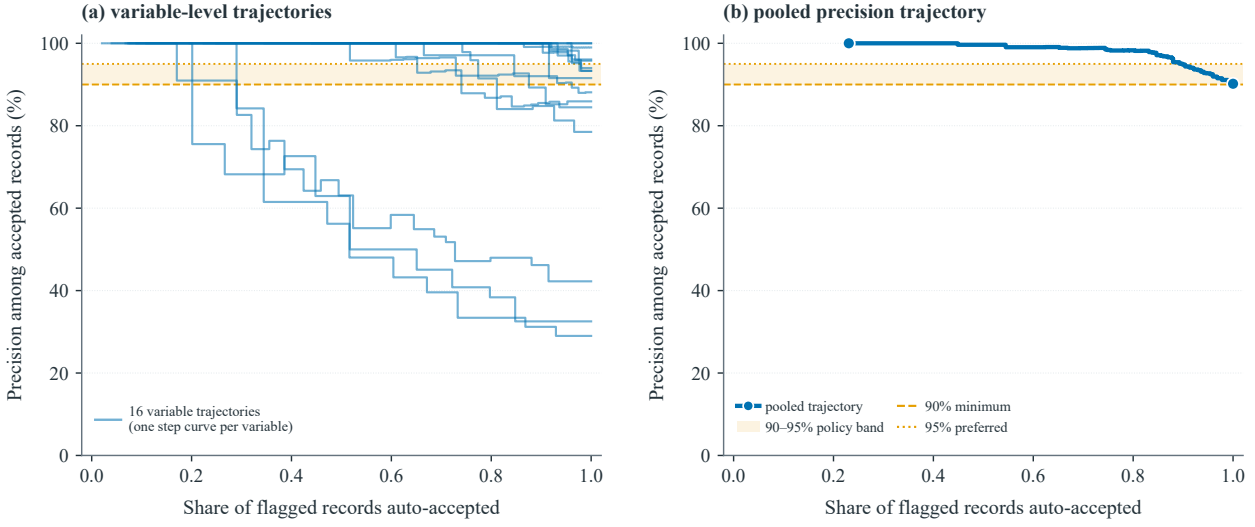


Figure C.2: Precision among accepted records against the share of flagged records auto-accepted, measured on the human reference set with the same construction as Figure 5b. (a) Each variable as a thin line, most holding 90% precision until more than four fifths of their flags are accepted. (b) All variables pooled, crossing 95% near nine tenths of the range and 90% only at its right-hand end. Both panels use the whole reference set, so they give the in-sample budgets 10.1% and 0.0%; Section 4.3 reports the cross-fitted budgets 10.0% and 0.8% instead.

D. Proofs and the implementation map

This appendix collects the mathematics on which the audit of Section 3.4 rests. It fixes the notation shared by every statement and proves that the Murphy decomposition of the Brier score is exact on the output grid. It then proves Proposition 1 and Lemma 2 together with a tightness statement for each, derives the closed form of the break-even fraction of Eq. (4), and closes with the map from every numbered equation to the released code. Every statement is written for the weighted empirical distribution of a labeled sample, with weights w_i that sum to n, and each holds verbatim for the population once weighted means are replaced by expectations.

Throughout, $G \subset [0,1]$ is the finite output grid, $W_v = \sum_i w_i \mathbb{1}[p_i = v]$ is the weight at grid value v, $r_v = \frac{1}{W_v}\sum_i w_i \mathbb{1}[p_i = v]\, y_i$ is the observed positive rate at v whenever $W_v > 0$, $\bar{p} = \sum_v (W_v/n)\, v$ is the mean probability, and $\bar{y} = \sum_v (W_v/n)\, r_v = \frac{1}{n}\sum_i w_i y_i$ is the base rate, written ρ in the main text. Grouping by grid value is exact rather than approximate, because every record at a value v carries the same probability. Two identities follow directly from the definitions,

$$\sum_i w_i \mathbb{1}[p_i = v]\,(y_i - r_v) = 0 \quad \text{for every } v, \qquad \sum_{v \in G} \frac{W_v}{n}\,(v - r_v) = \bar{p} - \bar{y}. \tag{D.1}$$

A model is calibrated on the sample when $r_v = v$ at every grid value with $W_v > 0$, and $w_0 = W_0/n$ denotes the weight the model places on the value zero.

Lemma D.1 (Exact Murphy decomposition on the grid). *With* BS *as in Eq. (10) and the grouping by grid value,*

$$\mathrm{BS} \;=\; \sum_{v \in G} \frac{W_v}{n}\,(v - r_v)^2 \;-\; \sum_{v \in G} \frac{W_v}{n}\,(r_v - \bar{y})^2 \;+\; \bar{y}(1 - \bar{y}), \tag{D.2}$$

with equality and no binning error, and the reliability term bounds the exact calibration error of Eq. (12) through $\mathrm{ECE}_G^2 \le \sum_v (W_v/n)(v - r_v)^2$.

Proof. Fix v with $W_v > 0$. For every record at v, $(v - y_i)^2 = (v - r_v)^2 + 2(v - r_v)(r_v - y_i) + (r_v - y_i)^2$. Summing with the weights w_i over the records at v, the middle term vanishes by the first identity of Eq. (D.1). Because $y_i \in \{0,1\}$ gives $y_i^2 = y_i$, the last term sums to $W_v r_v - W_v r_v^2 = W_v\, r_v(1 - r_v)$. Summing over v and dividing by n, $\mathrm{BS} = \sum_v (W_v/n)(v - r_v)^2 + \sum_v (W_v/n)\, r_v(1 - r_v)$. Expanding $\sum_v (W_v/n)(r_v - \bar{y})^2 = \sum_v (W_v/n)\, r_v^2 - \bar{y}^2$ and substituting gives $\sum_v (W_v/n)\, r_v(1 - r_v) = \bar{y}(1 - \bar{y}) - \sum_v (W_v/n)(r_v - \bar{y})^2$, which is Eq. (D.2). The inequality is Jensen's inequality for the square applied to the probability weights W_v/n and the values $|v - r_v|$.

Proof of Proposition 1. Suppose first that the model is calibrated on the sample, so that $r_v = v$ wherever $W_v > 0$. Then $\rho = \bar{y} = \sum_v (W_v/n)\, r_v = \sum_v (W_v/n)\, v = \bar{p}$, and because every positive grid value is at least δ, $\bar{p} \ge \delta \sum_{v>0} W_v/n = \delta(1 - w_0)$. Rearranging $\rho \ge \delta(1 - w_0)$ gives $w_0 \ge 1 - \rho/\delta$. The bound used only $v \ge \delta$, so it holds for every assignment of positive grid values, and the contrapositive is the first claim. If $\rho < \delta(1 - w_0)$, then $r_v = v$ fails at some grid value carrying weight, whatever values the model returns. For the second claim, the second identity of Eq. (D.1) and the triangle inequality give $\mathrm{ECE}_G = \sum_v (W_v/n)\,|v - r_v| \ge \left|\sum_v (W_v/n)(v - r_v)\right| = |\bar{p} - \rho| \ge \bar{p} - \rho$. The inequality $\bar{p} \ge \delta(1 - w_0)$ holds whether or not the model is calibrated, so $\bar{p} - \rho \ge \delta(1 - w_0) - \rho > 0$, which is Eq. (15). Equality in the triangle inequality holds exactly when $v - r_v$ has one sign at every grid value carrying weight, so $\mathrm{ECE}_G = \bar{p} - \rho$ for a model whose reliability curve lies on or above the diagonal everywhere, and the floor $\delta(1 - w_0) - \rho$ is attained when in addition all positive weight sits at δ. Finally, let $g : G \to [0,1]$ be any recalibration map. If $g(p)$ is calibrated then $\sum_v (W_v/n)\, g(v) = \rho < \delta(1 - w_0)$, so g must send some positive grid value carrying weight below δ. The maps of Eq. (20) can do so and the model cannot, which is why the floor is a property of the grid and not of the ranking.

Proof of Lemma 2. Write the coded field as $y' = y(1 - m)$, where m is an omission indicator with $\Pr(m = 1 \mid y = 1, p = v) = f_v$, so that $y' = 1$ implies $y = 1$ and $r'_v = \Pr(y' = 1 \mid p = v) = (1 - f_v)\, r_v$. The weights W_v depend on p alone and are therefore the same under both references. Since $0 \le f_v \le 1$, $r'_v \le r_v$, which is the first claim. If $(1 - f_v)\, r_v \ge v$ then $v \le r'_v \le r_v$, so $|v - r'_v| = r'_v - v \le r_v - v = |v - r_v|$, which is the second claim, and summing

with the weights W_v/n over every grid value with $W_v > 0$ gives Eq. (19). Under that condition the gap is exact rather than an inequality,

$$\mathrm{ECE}_G(y) - \mathrm{ECE}_G(y') \;=\; \sum_{v \in G} \frac{W_v}{n}\,(r_v - r'_v) \;=\; \sum_{v \in G} \frac{W_v}{n}\, f_v\, r_v \;=\; \Pr(y = 1,\; y' = 0), \tag{D.3}$$

the share of records that the narrative reference marks positive and the coded field does not, which is the narrative-only cell measured in Section 4.4. Where the condition fails at some v, $r'_v < v$ and $|v - r'_v| = v - r'_v$. This exceeds $|v - r_v|$ whenever $r_v < v$, and whenever $r_v \geq v$ with $2v > r_v + r'_v$, so omission then inflates the measured error and the bias reverses on that part of the grid. If the coded field also records factors the narrative does not state, r'_v can exceed r_v and the direction of the bias is no longer determined by the omission rate alone.

Lemma D.2 (Break-even fraction). *Under the cost model of Eq. (3) with $k_f > k_s$, the break-even fraction of Eq. (4) has the closed form*

$$q^* \;=\; \frac{\beta\,(k_f - k_s)}{\alpha + \beta k_f + b\bar{\ell}}, \qquad \frac{\partial q^*}{\partial \bar{\ell}} \;=\; -\,\frac{b\,\beta\,(k_f - k_s)}{(\alpha + \beta k_f + b\bar{\ell})^2}, \qquad \frac{\partial \ln q^*}{\partial \ln \bar{\ell}} \;=\; -\,\frac{b\bar{\ell}}{\alpha + \beta k_f + b\bar{\ell}}, \tag{D.4}$$

so q^* decreases in narrative length, and its elasticity with respect to length equals minus the share of the full-schema token count that the text contributes.

Proof. Screening costs less than coding everything once exactly when $T(k_s, \bar{\ell}) + q\,T(k_f, \bar{\ell}) < T(k_f, \bar{\ell})$, that is when $q < 1 - T(k_s, \bar{\ell})/T(k_f, \bar{\ell}) = [T(k_f, \bar{\ell}) - T(k_s, \bar{\ell})]/T(k_f, \bar{\ell})$. The intercept α and the text term $b\bar{\ell}$ cancel in the difference, which leaves $\beta(k_f - k_s)$ in the numerator. Differentiating the quotient in $\bar{\ell}$ gives the second expression, and multiplying it by $\bar{\ell}/q^*$ gives the third. When βk_f dominates $b\bar{\ell}$ the elasticity is small, which is the claim of Section 3.2 that the schema rather than the text governs the two-stage design.

The weighting of Eq. (5) is a calibration estimator rather than an inverse-probability one, and the distinction matters for what can be claimed about it. The classical result is that $\mathbb{E}\big[\sum_{i \in s} z_i/\pi_i\big] = \sum_{i \in U} z_i$ for any sample s drawn from the frame U with inclusion probabilities $\pi_i > 0$, because $\mathbb{E}\,\mathbb{1}[i \in s] = \pi_i$ (Horvitz and Thompson, 1952). That result is background here rather than an assertion about this reference set, because the greedy draw and the identifier screen of Section 3.3 leave no π_i that can be written down. The weights used instead satisfy the calibration equations exactly, so the weighted count of every cell reproduces its population count, and the resulting estimator is design-consistent under the condition that labeling is uninformative about the outcome within a cell (Deville and Särndal, 1992). The ratio of two weighted sums is consistent but not unbiased, with a bias of order $1/n$ (Särndal et al., 1992), which is why the percentile interval of Eq. (6) is re-centered by the bootstrap estimate of that bias. Table D.1 lists, for every numbered equation of Section 3, the released script, the function and the result file that compute it, so that a reader can locate the implementation of each quantity without searching the repository.

Data and code availability

The question schema, the asynchronous runner, the metric implementations with their unit tests, the figure and table generators, the dependency file and a release manifest naming what is public and what is not, and all aggregated outputs behind every figure and table are released at `https://github.com/pozapas/jev-calibrated-narrative-coding`. All runs use a pinned model identifier, recorded per call, and every script resolves its paths from the repository root.

The underlying narratives and the record-level files that carry crash identifiers are not redistributable under the data agreement, and the reference frame is among them, so it is not released. A de-identified pair-level audit table has been prepared instead, carrying an anonymous narrative identifier, the variable, the model probability, the consensus label, the dispute status, the calibration cell, the draw route, the narrative weight, the assignment probability, the pair weight and both frontier arms' predictions, with no crash identifier and no narrative text. It is the file from which the human-reference results of Section 4.3, the recalibration of Section 4.2 and the frontier comparison of Section 4.4 can be recomputed in full. It will be made available on request once the data owner has confirmed that the agreement permits it. The population-scale results of Section 4.1 and the downstream counts of Section 4.5 depend on the Stage-2

Table D.1
Where each formalized quantity is computed.

equation	quantity	script (`src/`)	function	result file (`data/`)
Eq. (1)	gating operator and no-match override	`s06_flatten.py`	`flatten`	`stage2_flat.parquet`
Eq. (2)	leakage rate λ_v	`s06_flatten.py`	`flatten`	`stage2_leakage.json`
Eq. (5)	Hájek ratio with calibration weights w_i	`s05b_design_weights.py`; `s17_gold_analysis.py`	`design_weights; wrate, block`	`gold_design_weights.csv`; `gold_analysis.json`
Eq. (6)	stratified cluster bootstrap interval	`s07_metrics.py`	`cluster_bootstrap; weighted_bootstrap`	`analysis.json`; `gold_analysis.json`
Eq. (10)	Brier score and Murphy decomposition	`s07_metrics.py`	`brier, brier_decomposition`	`analysis.json`
Eq. (11)	expected and maximum calibration error, equal-mass bins	`s07_metrics.py`	`ece, mce, equal_mass_bins`	`analysis.json`
Eq. (12)	exact calibration error on the output grid	`s07_metrics.py`	`ece_discrete, discrete_reliability`	`analysis.json`; `gold_analysis.json`
Eq. (13)	calibration slope and intercept	`s07_metrics.py`	`calibration_slope`	`analysis.json`; `gold_analysis.json`
Eq. (14)	Spiegelhalter statistic	`s07_metrics.py`	`spiegelhalter_z`	`analysis.json`; `gold_analysis.json`
Eq. (15)	resolution-floor bound	`s07_metrics.py`	`resolution_floor_report`	`analysis.json`
Eq. (17)	precision and coverage over flagged records	`s07_metrics.py`	`precision_coverage, coverage_at_precision`	`analysis.json`
Eq. (18)	review budget $H(\pi^*)$, cross-fitted	`s07_metrics.py`; `s17_gold_analysis.py`	`positive_review_budget; held_out_budget`	`analysis.json`; `gold_analysis.json`
Eq. (16)	risk–coverage curve and area	`s07_metrics.py`	`risk_coverage, aurc, coverage_at_risk`	`analysis.json`
Eq. (19)	one-sided reference-noise bound	`s17_gold_analysis.py`	`narrative_only`	`gold_analysis.json`
Eq. (3)	token cost model	`s08_cost_model.py`	`main`	`cost_model.json`
Eq. (4)	two-stage cost and break-even fraction	`s08_cost_model.py`; `pricing.py`	`main; jev_cost_per_1k`	`cost_model.json`
Eq. (20)	Platt and isotonic recalibration, split over narratives	`s17_gold_analysis.py`	`recalibration`	`gold_analysis.json`
Eq. (7)	Cohen's kappa, weighted against the reference set	`s07_metrics.py`; `s16_gold_ingest.py`	`cohen_kappa`	`gold_reliability.json`; `gold_analysis.json`
Eq. (8)	Clopper–Pearson interval	`s07_metrics.py`	`clopper_pearson`	`analysis.json`
Eq. (9)	McNemar test	`s07_metrics.py`	`mcnemar`	`analysis.json`

outputs joined to coded fields at crash level and cannot be recomputed from any releasable file, so the aggregated result files behind them are released in place of the inputs, and a controlled-access route through the data owner is the only way to reproduce them from source.

CRediT authorship contribution statement

Amir Rafe: Conceptualization, Methodology, Software, Formal analysis, Data curation, Validation, Visualization, Writing - original draft, Writing - review & editing. **Subasish Das:** Conceptualization, Methodology, Supervision, Writing - review & editing.

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